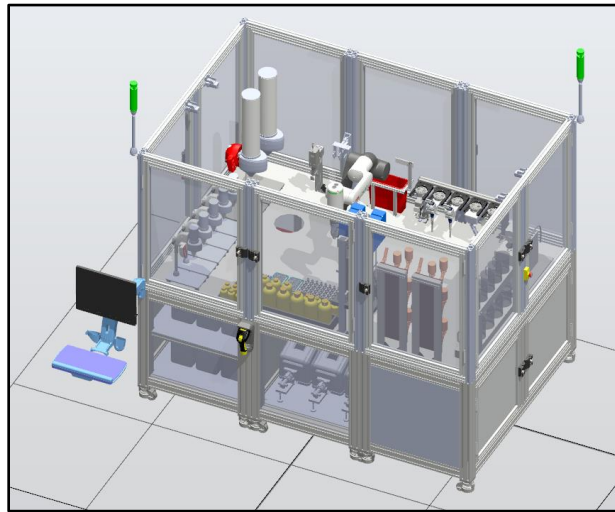




ABB Robotics LLC

Service Robotics



Prepared for

BASF

Robotic Formulation Preparation System

Q25-8248299 Rev. 0

3/25/2026

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3/25/2026

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Proposal Number: Q25-8248299 Rev. 0

Reference: Robotic Formulation Preparation System

Dear Vanathy Senthilnathan,

Thank you for contacting ABB for your automation needs. We are pleased to have the opportunity to provide you with this proposal regarding the purchase of ABB robotic equipment. With over 500,000 robots installed worldwide, ABB is committed to delivering quality robot products, software, and solutions designed to meet the most demanding robot automation requirements.

ABB wants you to know that your business is very important to us and we have made every effort to ensure that this offer is complete, competitive, and accurate.

ABB Robotics offers distinct products and advantages. Specifically, ABB makes our own positioners and will perform integration with the robots rather than sourcing from 3rd party companies, we have the longest mean time between failure of any robot in the industry and with more than 300 dedicated personnel ABB has best-in-class service footprint to further ensure machine up time. All these points underscore ABB's commitment to superior system solutions for our customers.

If you have any further questions or comments after reviewing this information, please contact me at the number or email listed below.

Sincerely,

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Table of Contents

1	Title of Project/Project Name Description	5
1.1	System Description	5
1.2	Design Components and Devices.....	6
1.3	System Layout.....	12
1.4	Automation Process	15
1.5	System Specifications.....	19
1.6	System Throughput and Capacity.....	23
1.7	System Software Architecture	Error! Bookmark not defined.
1.8	System Operator Interface.....	24
2	Investment Summary:.....	26
2.1	Option 1.....	26
2.2	Additional items	26
3	Ordering Instructions to the Buyer:	27
3.1	Payment Information	27
4	Conditions of Sales	28
4.1	Construction Location:	28
4.2	Conditions of Safety.....	28
4.3	Validity:.....	28
4.4	Freight:	28
4.5	Warranty:.....	28
4.6	Third Party Material Warranty:	28
4.7	Tariff Clause:	29
4.8	Market Conditions:	29
5	ABB's Responsibilities and Support Items	30
5.1	System Engineering.....	30
5.2	Project Schedule.....	30
5.3	Project Kick-off.....	30
5.4	Concept Design	31
5.5	Detailed Design	31
5.6	System Assembly	32
5.7	Pre-FAT Debug and Validation.....	33
5.8	Factory Acceptance Test (FAT).....	34
5.9	Site Acceptance Test (SAT)	36
5.10	Request for Changes – CORs.....	40
5.11	Communication.....	40
5.12	Documentation.....	40
5.13	Additional System Commissioning and Start-Up Support	41
5.14	Clarifications and Assumptions	41
6	Company Name Responsibilities:.....	43
7	ABB ROBOTICS LLC – ROBOTICS BUSINESS AREA US GENERAL TERMS AND CONDITIONS OF SALE.....	45

1 Robotic Formulation Preparation System Description

The proposed system is a flexible robotic platform designed to automate the preparation of material formulations through coordinated liquid and solid dosing, controlled mixing, and automated cleaning. Centered around a robot arm, the system integrates multiple dosing technologies to handle a wide range of material properties—from low-volume liquids to viscous fluids and bulk solids—within a single configurable workflow. Recipes, material sequencing, and process parameters are defined by the operator, allowing the system to adapt to evolving formulation needs while maintaining repeatability, traceability, and controlled execution of each preparation cycle.

1.1 System Description

The system is designed to prepare formulations consisting of both liquid and solid components using a combination of precision dosing technologies and robotic handling. Liquid materials are introduced through pipettes, syringe-based transfer for viscous fluids, and peristaltic pumps for larger volumes, while solid materials are dispensed using gravimetric dosing (for small quantities) and precise volumetric dosing systems (for larger quantities). All materials are combined within a mixing container, where intermediate and final mixing steps are performed according to the defined recipe. The system supports configurable sequencing, enabling the operator to define not only quantities but also the order of addition and any required mixing steps between additions.

A central robotic arm coordinates all operations, including material handling, tool manipulation, dosing interactions, and transfer of containers between system stations. Following formulation, the system performs an automated cleaning cycle of the mixing element to prepare for subsequent runs. Waste streams are managed through integrated solid and liquid waste handling systems, and all process steps are tracked through location-based identification. The system is designed to enhance consistency and process control while preserving operator flexibility and oversight. It does not alter or assume responsibility for analytical results, as testing methods remain external to this system.

1.2 Design Components and Devices

Table below Highlights key system hardware components. These components may change at design freeze but are currently in scope for what we intend to use for the integration of the system.

Item	Qty	Description of Scope	Picture
1	1x	Robotic Arm: GoFa™ CRB 15000 - 5Kg ABB to Supply as part of proposal Comments: The ABB GoFa robot serves as the central automation platform, executing all physical operations within the system. It performs container transfers, engages with dosing devices, manipulates tools such as pipettes and syringes, and positions containers for mixing and cleaning. The robot enables precise, repeatable execution of workflows while maintaining flexibility to accommodate different recipes and material types.	
		System Base Robotic Table ABB to Supply as part of proposal Comments: The automation table serves as the primary workstation, supporting all instruments, consumables, input/output locations, and safety features. It defines the robot's work envelope and ensures mechanical stability, repeatability, and safe operation. The cell top is open and designed to allow connection to an external ventilation system provided by BASF, such as building extraction, Hepa filtration, or other compatible solutions. The system includes predefined anchor points to facilitate integration; however, any additional modifications required will require a Change Order Request (COR). Final implementation of the extraction approach will be defined according to site-specific requirements. Size: 1600x2400x2000 mm (WxDxH)	
	1x	Robotic Grippers and Fingers: ABB to Supply as part of proposal Comments: The system uses task-specific end effectors to handle containers and tools. - Zimmer long-stroke grippers are used for Mixing containers - Zimmer long-stroke grippers are used for Sample containers - Intermediate vials These interchangeable tools allow the robot to switch between handling modes without manual intervention. Additional grippers required to accommodate new or different consumables can be integrated into the system through a Change Order Request (COR) process.	
	2x	Pipette Jig and Robotic Interface ABB to Supply as part of proposal	

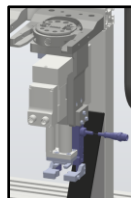
		Comments: Custom-designed pipette jigs and adapters provide the mechanical interface between the robot and the pipettes. These fixtures ensure proper alignment, secure mounting, and consistent positioning during operation. The jig enables the robotic system to reliably manipulate and actuate the pipettes, maintaining dosing accuracy and repeatability across all liquid handling steps.	
	1x	Syringe Gripper ABB To Supply as part of proposal Comments: The syringe holder end effector is a mechanical gripping tool used to securely hold and actuate a syringe during automated liquid transfer operations. It provides controlled positioning of the syringe barrel and plunger interface, allowing precise mechanical movement of the plunger while maintaining stable alignment with source and destination containers. The end effector itself does not define liquid accuracy; it enables mechanical control while the dispensed mass is verified gravimetrically by weighing the syringe before and after the dispense. This tool is applied in liquid handling scenarios where pipetting is not appropriate, such as viscous fluids, volatile solvents, or chemically aggressive media. It supports container-to-container liquid transfer under gravimetric control, with delivery volume determined by plunger displacement. Correlation between plunger movement and dispensed volume is established by the end user for the specific sample types used, while the system controls only plunger travel distance. The system requires the liquid level to remain at a height that allows the syringe needle to reach it from above while the bottle remains upright; this is to be ensured by the end user.	
	5x	Automatic Tool Changer (ATC) Station ABB To Supply as part of proposal Comments: An Automatic Tool Changer (ATC) station is integrated within the system to store and manage all robotic end effectors and grippers used throughout the workflows. The station provides defined docking positions for each tool, allowing the robot arm to automatically pick up and return grippers as required for handling different materials. The ATC ensures repeatable mechanical alignment and secure coupling of tools, supporting reliable operation across varied process steps while minimizing downtime and operator involvement. Tool presence and engagement status are monitored through the automation control system to maintain workflow continuity and operational safety.	
		B&R Automation PC 4100 ABB to Supply as part of proposal Includes one PC for system control and user interface. Used for workflow configuration, process monitoring, and data export.	
		Safety system	

		ABB to Supply as part of proposal A preliminary risk assessment has determined that the cell must be enclosed with hard walls and equipped with safety interlocks. The enclosure will include 10 access doors, which will remain automatically locked whenever the robot is in operation. To gain entry, the operator will be required to request access through the HMI, which will safely disable robot motion and release the corresponding door interlock.	
	1x	Mettler Toledo Analytical Balance: BASF To Supply as part of proposal Comments: For solid dosing below 6g, the system uses a Mettler Toledo gravimetric dosing scale. Materials are dispensed into an intermediate vial, which is subsequently transferred by the robot to the mixing container. An optional rinsing step—performed up to three times—ensures complete material transfer, with the rinsing solvent defined by the operator. Prior to each rinse, the robot gently agitates the vial to improve recovery. The system supports automated exchange of dosing heads, while the operator is responsible for loading dosing heads and defining and assigning their contents within the system.	
	1x	Stability base for analytical Balance BASF To Supply as part of proposal Comment: This base is to ensure weighing accuracy is maintained within the robotics system. Speirs Robertson AMB-30x45 AMB Balance Platform, Passive Air Isolators; 30x45 cm	
	7x	XQ Solid Dosing Systems BASF To Supply as part of proposal Comments: For solid dosing above 6g, the system uses XQ solid dosing devices to dispense materials directly into the mixing container. The system includes a total of 7 XQ dosing devices, consisting of 2 x SDB-2 units and 5 x SDB-1 units. This arrangement allows multiple solid materials to be available on the system at one time and supports flexible recipe execution across a range of formulation requirements. The operator is responsible for loading the dosing heads onto the XQ devices and identifying which material is assigned to each position through the barcode system described in section 1.4. Based on this configuration, the robotic system can select the appropriate XQ device and execute the required dosing step as part of the automated workflow. Direct dosing into the mixing container reduces intermediate handling steps and supports efficient transfer of larger solid quantities into the formulation process.	
	2x	Mettler Toledo/ Rainin E4 Electronic Pipette: BASF To Supply as part of proposal Comments: The system incorporates two Rainin pipettes to enable precise liquid handling across different volume ranges. Each pipette is selected based on the required transfer volume and application, allowing the system to handle both small and	

		moderate liquid additions within a single workflow. The robotic system interacts with the pipettes to perform controlled aspiration and dispensing of liquids, supporting accurate and repeatable dosing as part of the formulation process.	
	3x	MasterFlex Peristaltic Pump BASF To Supply as part of proposal Comments: The system incorporates three peristaltic pumps for transferring larger volumes of liquid. These pumps are connected to an automated valve, allowing controlled dispensing of liquids into the formulation container. Pumps can be assigned to specific materials or reconfigured by the operator, providing flexibility in handling bulk liquid inputs while maintaining controlled delivery. Each pump can dispense up to 4 liquids, depending on purging priming requirements (one liquid might be used for purging and priming tubes).	
		Mixing System (IKA Mixer) BASF To Supply as part of proposal Comments: The system utilizes an IKA mixing device equipped with an appropriate mixing blade to homogenize formulations following material dosing. The robotic arm transfers the mixing container to a dedicated staging interface that aligns and secures the container beneath the mixer. The container gripping and support mechanism is designed to maintain stability during mixing while avoiding excessive force that could deform the container. Final grip force and stability will be validated during system testing using representative materials to ensure reliable operation without container slippage or damage. Mixing parameters, including speed, are defined by the operator within the recipe to accommodate different material properties and to minimize undesirable effects such as air incorporation. Following each formulation, the system performs an automated cleaning cycle of the mixing blade. The robot prepares a dedicated cleaning container by dispensing washing solution (via the pump system, water or other cleaning solutions, assigned by the operator), positions it under the mixer, and runs the mixer at a low speed to clean the blade. After cleaning, the robot transfers the container and decants the solution into the liquid waste system. The effectiveness of the cleaning process or quality control requirements remain the responsibility of the end user, and are not validated by ABB. To manage residual drips from the mixing blade, a secondary collection container is placed beneath the mixer by the operator and must be periodically replaced between runs.	
		Waste Management System ABB to Supply as part of proposal Comments: The system incorporates dedicated waste handling for both solid and liquid streams, designed to safely collect, contain, and manage byproducts generated during automated formulation. Waste handling is integrated into the	

		workflow while maintaining clear operator responsibilities for monitoring and disposal. Solid Waste Management: Solid waste is collected in designated containers within the system workspace. This includes a dedicated sharps container for disposal of used pipette tips and syringes, as well as a separate container for vials, whether empty or containing residual material, generated during dosing and transfer steps. These containers are positioned for both robotic and operator access, ensuring efficient removal of waste from the process area. Some waste items may contain residual liquids; therefore, all handling and disposal must follow applicable site safety procedures. The operator is responsible for regularly monitoring, removing, and replacing solid waste containers as needed to maintain safe and continuous system operation. Liquid Waste Management (Sink and Collection System): Liquid waste generated during operation—primarily from cleaning cycles, rinsing steps, and decanting—is directed to an integrated sink within the system. The sink includes a flushing mechanism to rinse residues after disposal events, helping to prevent buildup and cross-contamination between runs. All liquid waste is routed from the sink into a dedicated waste collection container. Connection to building sewage is possible and would be in the scope of BASF. Liquid Waste Monitoring and Operator Responsibility: The liquid waste container is equipped with level sensors to monitor fill status during operation. The system provides alerts through the user interface when the container approaches capacity, allowing the operator to intervene before overflow conditions occur. The operator is responsible for emptying and maintaining the liquid waste container in accordance with facility procedures. The system does not perform automatic waste disposal beyond internal collection and routing.	
	1x	Barcode Scanner ABB to Supply as part of proposal The barcode scanning system enables traceability of all samples and formulation process. Barcodes are read to associate each container, material, and preparation stage with its digital record. The system links barcode identity with physical location inside the automation platform, allowing full tracking of where each sample is loaded. This supports electronic batch documentation and error prevention. Barcode data is integrated into the system logs and can be transferred to external platforms such as LIMS.	
		Input and Output Area ABB to Supply as part of proposal	

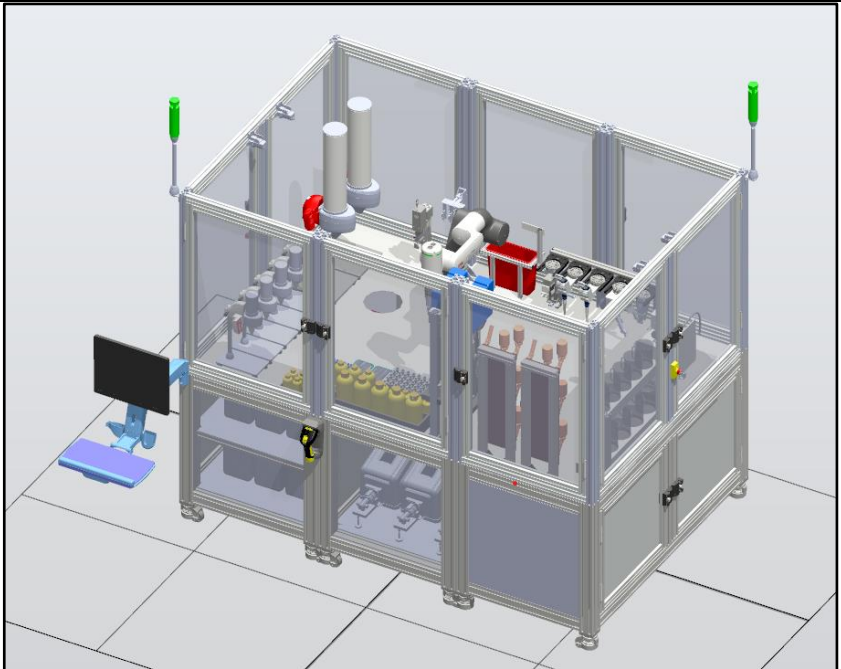
		Dedicated zones for operator loading of consumables and retrieval of finished materials, designed to maintain process segregation and workflow clarity. • Rack of 20 MT dosing heads • Rack of 10 100ml bottles (yellow) • Rack of 5 250ml bottles (yellow) • Rack of 5 500ml bottles (yellow) • Rack of 5 1000ml bottles (yellow) • 3 x Pipette tips boxes locator • Rack of 5 1L mixing vials • Rack of 5 500ML mixing vials Comments: The system is designed to accommodate the consumables and sample racks listed. Significant changes to these items may impact the system design, tooling, and workflow, and may require a Change Order Request (COR).	
	1x	Labx License BASF To Supply as part of proposal Comments: License for all included Mettler Toledo instruments (scales, pipettes)	
		Optifact: ABB to Supply as part of proposal Comment: OptiFact™ integrates directly with LIMS, enabling synchronized lab-to-line data flow. This connectivity enhances traceability, accelerates issue resolution, and supports data-driven decision-making across the production cell.	

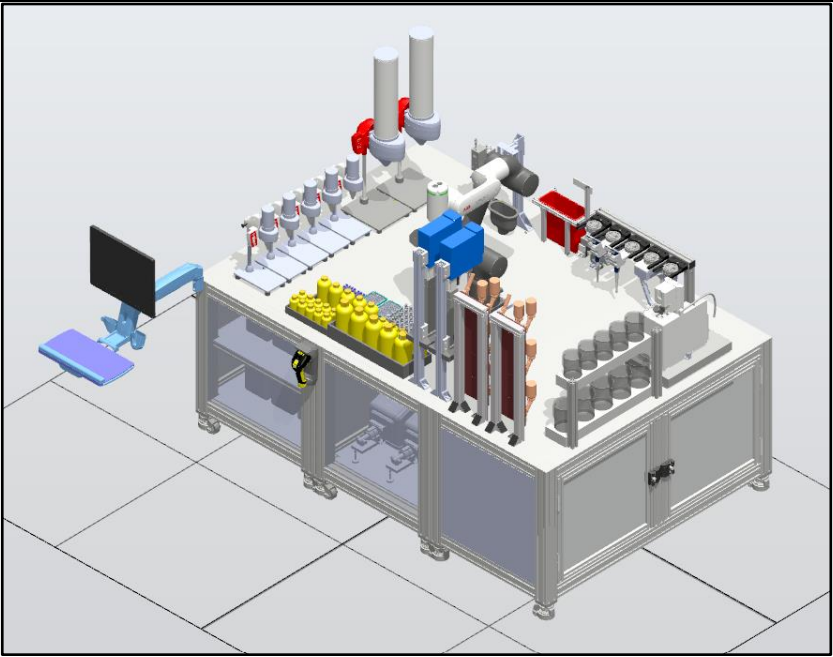
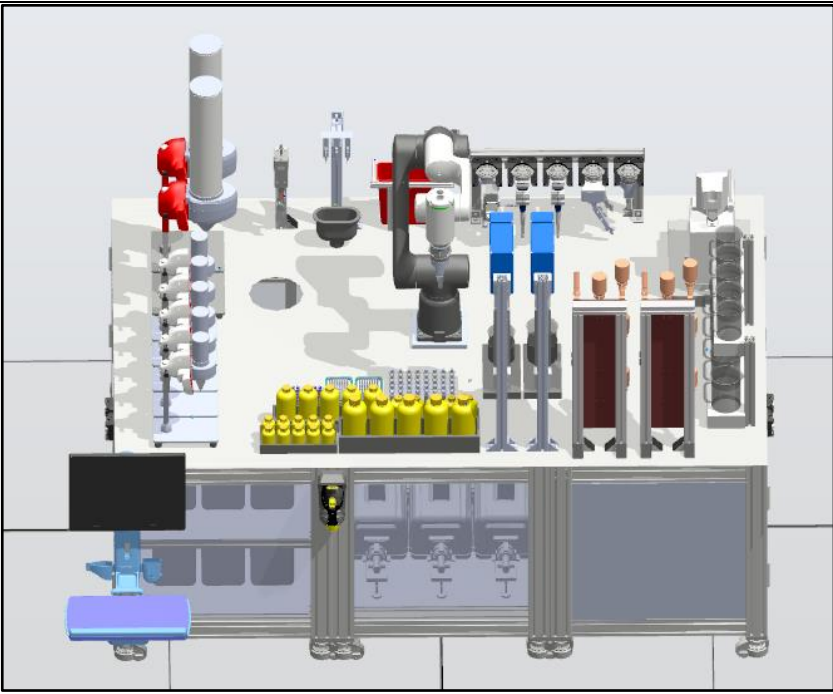
1.3 System Layout

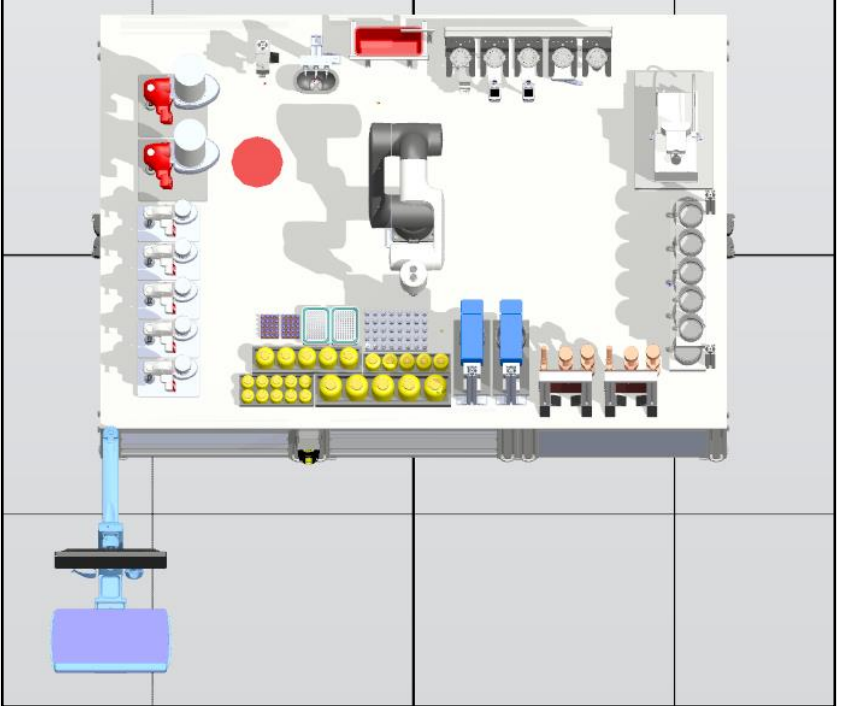
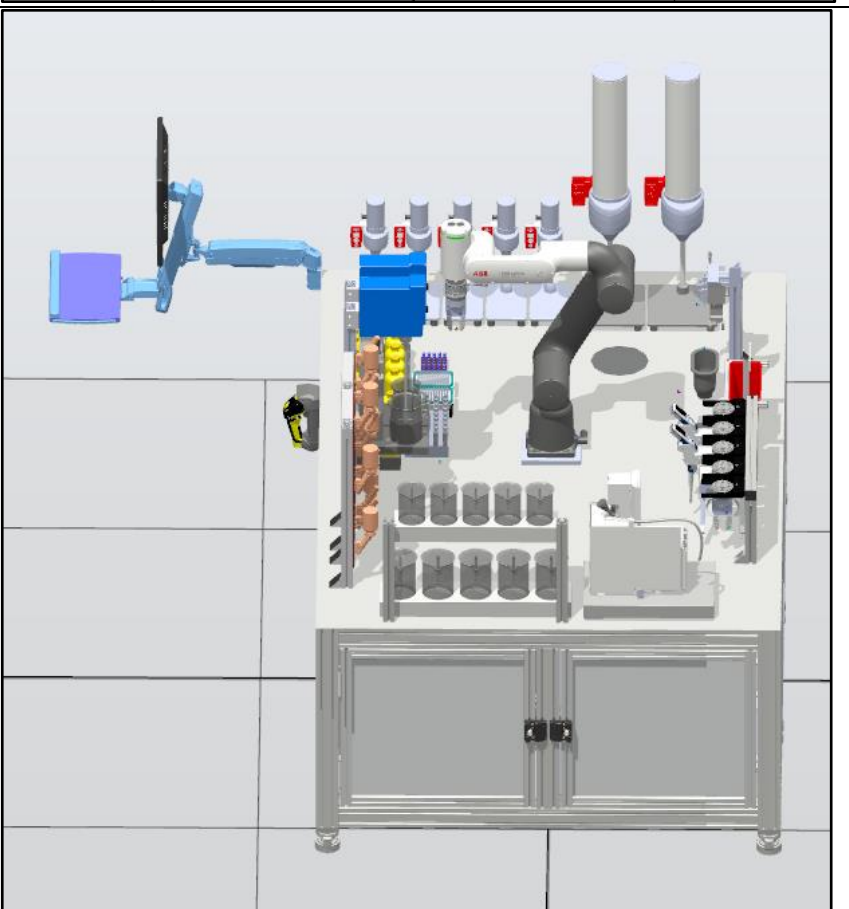
This section provides a preliminary visual representation of the system layout, showing the relative positioning of major devices, the robotic arm, input/output areas, and supporting infrastructure. The purpose of the layout is to illustrate the overall design concept and system footprint.

The layout is intended to help stakeholders visualize how the system components will integrate in a laboratory or production environment. It should be viewed as an initial design concept rather than a finalized configuration. Detailed engineering drawings and final dimensions will be developed during the detailed design phase and may differ from what is presented here.

To ensure clarity, it is important to note that this layout is provided for conceptual purposes only. The design shown here is not binding and is subject to change as project requirements evolve, facility constraints are confirmed, and engineering validation progresses. Drawings may not be to scale and should not be used for construction, facility modifications, or space allocation without further verification.

Item	Description	Picture
1	Full layout with guarding	

2	Layout without guarding for better view	
3	Front view of the cell without guarding better view	

4	Top view of the cell without guarding better view	 A 3D CAD rendering showing a top-down perspective of a robotic work cell. The cell is a white rectangular cabinet with various components mounted on top. On the left, there are two red cylindrical components. In the center, a black robotic arm is positioned. To the right, there are several blue and orange components. A red circle is drawn on the top surface of the cabinet. The cell is mounted on a grey base.
5	Side view of the cell without guarding better view	 A 3D CAD rendering showing a side view of the robotic work cell. The cell is a white rectangular cabinet with various components mounted on top. On the left, a blue robotic arm is visible. In the center, a black robotic arm is positioned. To the right, there are several blue and orange components. The cell is mounted on a grey base.

1.4 Automation Process

The system operates based on operator-defined recipes that specify materials, quantities, sequencing, and mixing steps. The workflow begins with the operator loading all required materials and consumables, assigning materials to dosing devices, and configuring the formulation either manually through the user interface or via file upload.

Once initiated, the robotic system executes the formulation sequence by selecting the appropriate dosing method for each material. Liquids are transferred using pipettes, syringes, or pumps depending on volume and viscosity, while solids are dispensed using either gravimetric or volumetric dosing systems. A maximum of up to 20 materials (mixtures), depending on the materials available on the system, can be used within a single formulation process. Intermediate mixing steps may be performed throughout the process as defined in the recipe.

After all materials have been added, the system performs final mixing and presents the completed formulation in output. A cleaning cycle of the mixing element is then automatically executed to prepare for the next run. The workflow supports operator intervention, allowing additional materials to be introduced if required, with all interruptions recorded for traceability. The system maintains a complete digital record of each formulation, including material usage, process steps, and any deviations or manual interactions.

Graphics below briefly describe the process summary showing key process steps. Consumables, equipment, tools, or other objects mentioned are non-binding and are for reference only. The process steps may be subject to change at final engineering phase of the project subject to COR for major changes.

The operator is responsible for physically loading the chemicals into the system and registering them with the robot. This requires two steps:

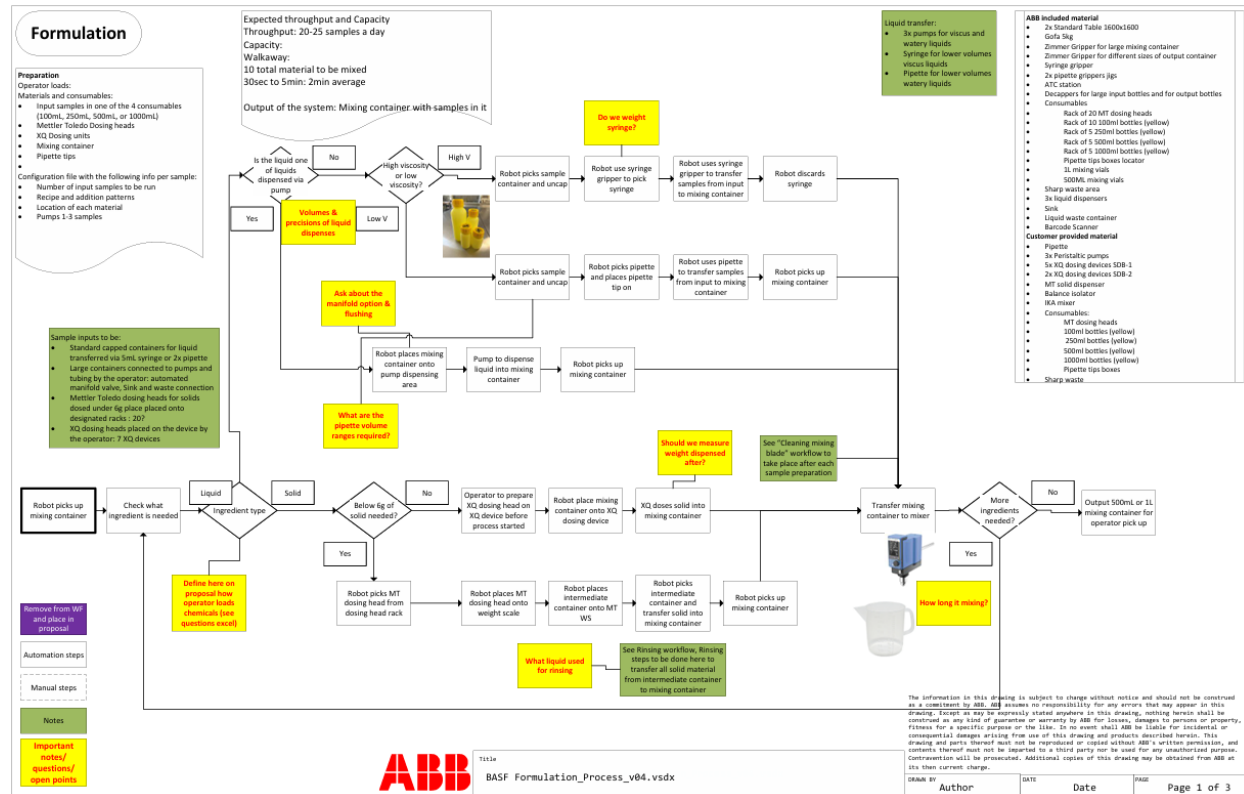
1. Scanning the barcode of the system position and the barcode of the chemical to pair them directly
2. Uploading a CSV file containing the mapping between barcodes and chemicals.
This database can be a standard reference list or updated dynamically whenever new chemicals are introduced.

The database also stores additional information for each chemical, including physical properties (liquid or solid), container type and size, and the appropriate handling method (pipette, syringe, pump, gravimetric solid dispenser, volumetric solid dispenser). The system uses this information during execution as instructed by the recipe or workflow, as long as the data matches the database.

If any discrepancies are detected, such as mismatched barcodes, incorrect container types, or incompatible handling instructions, the system will flag a warning and pause in a safe holding state until the issue is resolved by the operator.

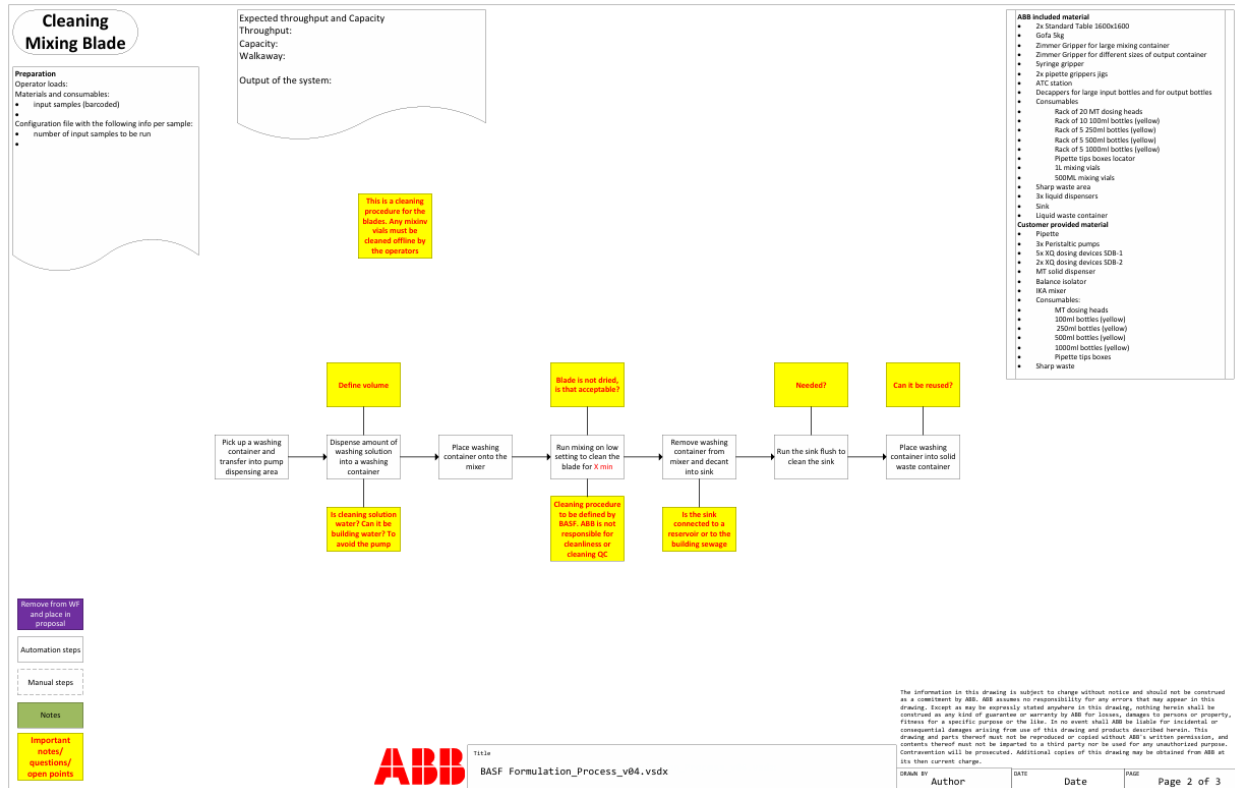
The system requires access to a facility drain for disposal of liquid waste generated during operation, including cleaning steps. A suitable sewer connection must be provided by the end user to enable proper routing and disposal of liquid waste.

1.4.1 Formulation Workflow



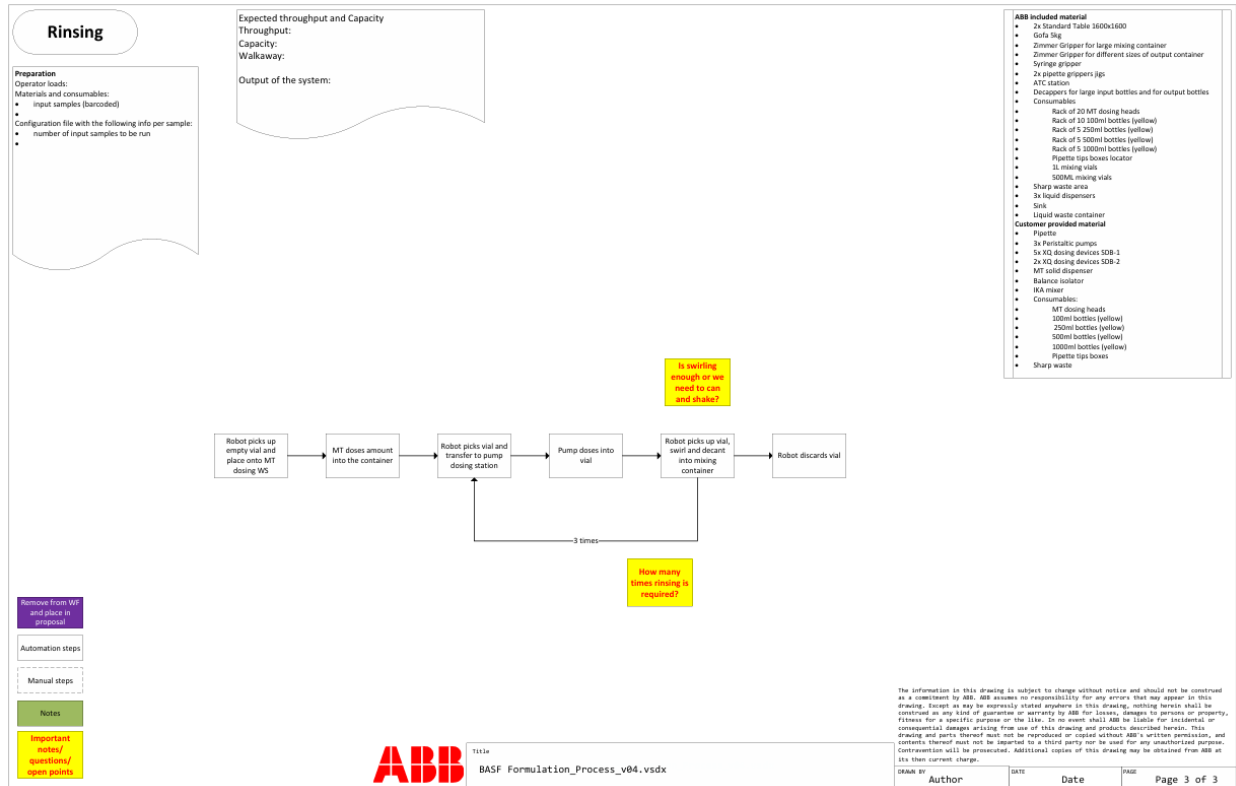


1.4.2 Cleaning Mixing Blade





1.4.3 Rinsing workflow



1.5 System Specifications

Item	Description	Picture
1	1L Mixing/reaction container - Loaded by the operator into the designated input/output area empty - Picked by the robot using the dedicated gripper - Used as the primary vessel for material accumulation, mixing, and final formulation output - Positioned by the robot onto the mixing station - Receives liquids and solids through various dosing methods - Remains uncapped during processing; no automated capping/decapping - Used for both intermediate and final mixing steps as defined in the recipe - After completion, remains in the output area for operator pickup - Assigned batch number relative to the output location - System has capacity to hold 5x1L reaction container at once	
2	500mL Mixing/reaction container - Loaded by the operator into the designated input/output area empty - Picked by the robot using the dedicated gripper - Used as the primary vessel for material accumulation, mixing, and final formulation output - Positioned by the robot onto the mixing station - Receives liquids and solids through various dosing methods - Remains uncapped during processing; no automated capping/decapping - Used for both intermediate and final mixing steps as defined in the recipe - After completion, remains in the output area for operator pickup - Assigned batch number relative to the output location - System has capacity to hold 5x500mL reaction container at once	

3	Sample/solution container 100mL, 250mL, 500mL, and 1000mL • Loaded by the operator into designated input positions • Containers are assumed to be pre-filled • The operator is responsible for identifying the contents of each container by either selecting the material from an existing system library or entering the information through the user interface, including the appropriate method of transfer. The operator must also assign and confirm the container's location within the system to ensure proper tracking and traceability throughout the process. • Robot performs controlled inversion (3–5 times) and agitation prior to uncapping and sampling to ensure homogeneity • Accessed by pipette or syringe depending on material properties and volume requirements • Liquid level must be maintained within operational limits to ensure proper access • System has the capacity to hold 10x100mL, 5x250mL, 5,500mL, 5x1L at once	
	Pipette tips Accounted for 2 different pipettes • Loaded by the operator into designated tip positions within the system • Picked up automatically by the pipette during liquid handling operations • Replaced between transfers to prevent cross-contamination • Disposed of automatically by the robot into the sharps waste container after use • Single-use consumables; no reuse within the system • System has the capacity to hold 2 pipette tip boxes at once	

	Syringes Syringe and needle • Syringe and needles are preassembled by the operator before entering the system • Loaded by the operator into designated staging locations • Picked and manipulated by the robot using the syringe gripper • Used for aspiration and dispensing of viscous liquids • Single-use consumables; no reuse within the system • After use, disposed of into the sharps waste container	
	MT dosing heads • Loaded by the operator onto a dosing head rack (up to capacity 20) • Each dosing head is assigned to a specific solid material and identified within the system by operator by its location on the rack during loading • Automatically picked, placed, and exchanged by the robotic system onto the dosing scale • Used for gravimetric dispensing of solids (≤ 6 g) into intermediate vials • Returned to designated rack position after use	
	Sharps Waste Bin • Positioned within the system and accessible to the robot for automated disposal • Used for collection of used pipette tips and syringes • Remains in place throughout operation; no robotic handling of the container itself • Monitored and replaced by the operator as required in accordance with site safety procedures	

	Intermediate Vials • Loaded by the operator into designated vial racks within the system • Used as temporary containers for solid dosing from the Mettler Toledo system (≤ 6 g) • Receive accurately dispensed solid material from the MT dosing scale • Picked and transferred by the robot to the mixing container for material delivery by decanting • Used in conjunction with rinsing steps (up to 3 times) to ensure full transfer of solids into the mixing container • During rinsing, liquid is added to the vial and the robot performs gentle agitation to recover residual material before transfer • Remain uncapped during operation; no automated capping/decapping included • After use, vials may contain residual material or liquid and are transferred to a designated waste container	
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1.6 System Throughput and Capacity

Preliminary numbers of throughput and capacity have been calculated based on the process timings and robot motions described in the workflows above. These numbers are non-binding and are subject to change upon a closer look during detailed engineering and can be impacted if there are changes such as added steps to the workflow, consumable modifications, changes to target tolerances, etc...

System throughput is highly dependent on which dosing methods and samples are processed. Throughput below is considering different scenarios resulting in different cycle times.

Two minutes mixing happens between each addition,

Series	Workflow	Cycle time	System Capacity	Throughput
1	<6g mixing: 1 >6g mixing: 1 Pump dosing: 5 Pipette dosing: 2 Syringe mixing: 1 Total of 10	57 minutes per formulation	Up to 5x500mL and 5x1L formulation.	24 samples / day
2	<6g mixing: 1 >6g mixing: 2 Pump dosing: 7 Pipette dosing: 2 Syringe mixing: 1 Total of 13	67 minutes per formulation	Up to 5x500mL and 5x1L formulation.	21 samples / day
3	<6g mixing: 3 >6g mixing: 1 Pump dosing: 4 Pipette dosing: 1 Total of 13	59 minutes per formulation	Up to 5x500mL and 5x1L formulation.	24 samples / day

1.7 System Operator Interface

The user interface is designed to provide intuitive, flexible control over the system while supporting full process transparency and traceability. Operators can configure workflows by selecting materials, defining quantities, sequencing additions, and specifying mixing steps. Recipes can be created directly within the interface or uploaded via structured files such as CSV.

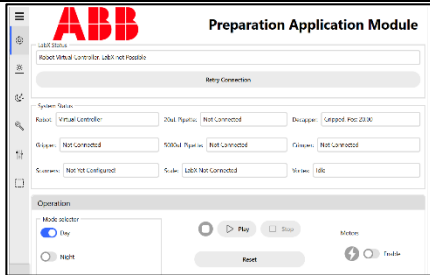
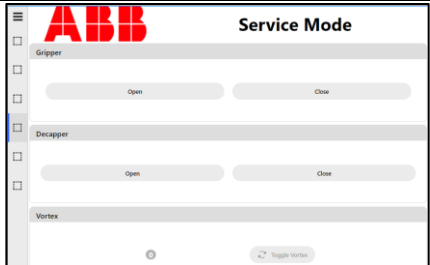
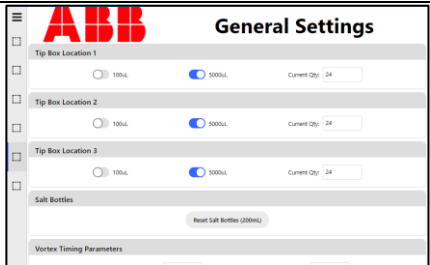
The system integrates barcode tracking and location-based logic to maintain traceability of all materials and samples throughout the process. Real-time system status is displayed, including active operations, workflow progress, and system alerts such as waste capacity or process interruptions.

For each processed sample, the system generates detailed log files capturing the full execution history, including material additions, timestamps, and operator interactions. These records are accessible and exportable, supporting both operational review and data traceability requirements.

ABB HMI Screen examples are non-binding and are for reference.

Item	Description	Picture
1	UI is used for operation of the system: contains images on the setup of the cell. toggles and configuration for inputs. allows inputs to be configured with a file. Shows status of all the systems.	
	Daytime Mode, Configure Workflow parameters	
2	Nighttime mode, configure workflow and sample parameters.	



3	Control robot operation and view status of system components, Including LabX integration.	
4	Service mode for maintenance or troubleshooting	
5	General Consumable settings and tracking. Allows for	

2 Investment Summary:

2.1 Option 1

Item	Description Option A	Total in USD
1	Robotic Formulation Preparation System: ➤ Hardware supplied by ABB for purpose of integration and functionality. ➤ Section 1.2, 1.3, 1.4 describe the scope of system	
2	Robotic Formulation Preparation System: Time and Labor ➤ Design and development ➤ System Assembly ➤ Mechanical and Electrical Integration ➤ Commissioning ➤ FAT and SAT Testing Section 5 ABB Responsibilities	
3	Production Ready Workflow	
	a. Workflow Part 1	
	b. Workflow Part 2	
	Total Project Cost	\$499,985

Table 1 Note:

2.2 Additional items

Item	Description Option A	Total in USD
1	➤ Post SAT additional Onsite Support	\$10,300
2	➤ 1x Training Certificate	\$3,100

Table 2 Note: additional support weeks do not include travel time or T&E that will be charged at cost +10% processing fee. Training certificates can be purchased together with the project or directly from [ABB university website](#). Training price is subject to be updated by ABB and will be posted there

3 Ordering Instructions to the Buyer:

- Please reference quote Q25-8248299 Rev. 0
- Payment terms of: **Net 30**
Invoices will be rendered and become due for payment in accordance with the following schedule:
 1. **25% Down Payment with Purchase Order**
 2. **30% Layout/Design Approval**
 3. **35% FAT approval at ABB Auburn Hills, MI Facility**
 4. **10% SAT approval** [not to exceed 60 days from shipment]
- Incoterms: Freight FCA Auburn Hills, MI USA, Freight Collect
- ABB Terms and Conditions apply and must be referenced in the Purchase Order.
- If applicable, please include tax exemption certificate with order.
- E-mail P.O. to: RPOrderInbox@us.abb.com

3.1 Payment Information

ABB Robotics LLC Tax ID: 39-3366449 Mail Payment To: ABB Robotics LLC PO Box 88868 Chicago, IL 60695-1868	ACH/Wire Transfer to: ABB Robotics LLC Citibank, N.A. Delaware Branch ABA Number: 031100209 SWIFT Code: CITIUS33DEL Account No.: 54346509
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4 Conditions of Sales

All prices quoted in this proposal are in US dollars. Exclusive of all federal, state, municipal or other government excise, sales, use, occupational or like taxes, (including import and export duties) imposed by either the country of manufacture or the receiving country, not in force or to be enacted in the future.

4.1 Construction Location:

Construction and run-off of the system will be executed at Auburn Hills, MI.

A credit history review will be conducted upon receipt of a purchase order and must be approved by the ABB N.A. Executive Committee prior to acceptance of a purchase order.

Final payment terms may be modified upon results of the credit review.

4.2 Conditions of Safety

Additional terms and conditions are the attached ABB Robotics LLC Conditions of Sale.

4.3 Validity:

This quotation is valid for 90 days.

4.4 Freight:

The equipment included in this proposal shall be shipped F.O.B. Auburn Hills, Michigan and is freight collect.

4.5 Warranty:

Standard ABB Warranty is 12 months from start of production not to exceed 18 months from shipment, whichever occurs first, if an ABB Robot Installation Report is generated and submitted to ABB Customer Service Department. Any special warranty offering may be found in the quoted robot content description.

4.6 Third Party Material Warranty:

It is in the scope of this proposal that BASF will procure laboratory equipment needed and supply to ABB for the integration project. BASF is responsible for sourcing the equipment, devices, software, etc., with the correct specifications and supply to ABB within the deadlines specified by the project plan for the project to stay on schedule.

- Third party material where ABB selects and the vendor and supply equipment, ABB Warranty applies.
- Third party material where BASF directs ABB to specific vendor, that vendor's warranty applies. ABB will provide applicable warranty information to BASF.

4.7 Tariff Clause:

The price does not include any amounts for changes in taxes, tariffs, or other similar charges that may cause an increase in the cost of materials to be supplied or used to supply the equipment herein. ABB shall be entitled to an equitable adjustment in time and money for any costs that it incurs directly or indirectly that arise out of or relate to changes in taxes, tariffs, or similar charges due to such changes including, without limitation, escalation, delay damages, costs to re-procure, costs to change suppliers, costs of manufactured equipment or goods, or other costs of any kind resulting from the changes.

If during the contract period there are changes in laws, orders, regulations, tariffs, taxes, standard or bylaws, which affect the cost or time of performance of this contract, then an equitable adjustment shall be made to the date(s) and price(s) resulting therefrom.

4.8 Market Conditions:

Both parties are aware of the global supply chain and related challenges and uncertainties from outbreaks, pandemics, epidemics, war (declared or undeclared), government regulations and acts of governmental authorities (including sanctions), civil unrest, and general shortages of electronic components and availability and cost of other raw materials that may impact the cost of both parties and/or delivery schedule(s)/dates. Notwithstanding anything in this proposal to the contrary, the parties agree that in the event of the occurrence of any of the above aforementioned challenges, in the spirit of cooperation they will work together in good faith to make an equitable adjustment to the delivery dates, pricing and/or possible reductions of the contractually owed quantity of the equipment to be delivered to the purchaser, with a view to employing reasonable efforts to ensure that this proposal can be at least fulfilled in part. In the event the parties are unable to reach a mutually acceptable equitable adjustment within fifteen (15) working days for any of the foregoing, then the parties shall be excused from performance under the proposal which shall terminate. Each party waives any claim against the other party either for direct damages and/or loss of profits and/or indirect and/or intermediate and/or consequential and/or punitive damages, penalties and/or liquidated damages arising from or anyhow connected with the excusable events provided hereto.

Overtime, exchange rates fluctuate and may impair a party's ability to operate under the terms of this proposal. In such event the parties will meet to determine if there is a need for an equitable adjustment.

5 ABB's Responsibilities and Support Items

5.1 System Engineering

Every system order is assigned a project manager, who follows the job through start-up at the BASF's plant as the project is proposed. He or she is responsible for technical and contractual project management, project schedules, customer liaison, and coordination of design, build, assembly, and function test activities.

The system engineering function also produces system-related documents such as cell layouts, interface schematics, etc.

5.2 Project Schedule

A project management schedule will be provided showing BASF' milestones that would be covered for execution of the project. This form is initially broad in scope and will be made more detailed as the project progresses. This form will show both ABB's and BASF milestones and will be distributed at least once a month during the duration of the project. **All customer supplied material must be delivered to ABB within 3 weeks of PO to avoid impacts to the schedule.** Customer driven delays affecting key milestones, such as concept freeze or detailed design freeze, may require schedule extensions and could result in a Change Order Request (COR) to account for the additional time and resources needed.

KEY-MILESTONES	ABB	BASF	Phase Start (Week)
Project External Kick-off	x	x	0
Concept Engineering Phase	x	x	1
Detailed Engineering Phase	x		5
Manufacturing	x	x	9
Commissioning	x		19
Factory Acceptance (FAT)	x	x	25
Contract Shipment to BASF		x	26
Site Acceptance Tests (SAT)	x	x	29
Updated project timeline provided at project kick-off.			

5.3 Project Kick-off

During the project planning phase, ABB will finalize internal preparations, ensuring that engineering designs, procurement, and production schedules are aligned to the

information provided in this proposal before the project is presented to BASF. A detailed schedule, including key milestones and specific timelines, will be set to ensure transparency. Any potential risks will be thoroughly assessed and addressed upfront, with proactive measures in place to mitigate their impact. Information gaps should be noted and presented by both ABB and BASF during this call to assure proper alignment.

The project plan will be based on the information provided by BASF, with any gaps or additional requirements identified during the external kick-off addressed through a formal change order process. Any revisions to the scope may impact the cost, timeline, or deliverables, which will be mutually agreed upon before proceeding. The final project scope will be confirmed during the external kick-off, ensuring all parties are aligned before execution begins.

5.4 Concept Design

During the Concept Engineering phase, the project team reviews and refines the solution concept originally outlined in the proposal. Both ABB and BASF collaboratively examine the system workflows, hardware selections, and integration assumptions to ensure full alignment and close any outstanding gaps. This phase focuses on validating feasibility, clarifying requirements, and confirming that the proposed approach meets the customer's operational needs.

As part of this effort, ABB will develop an enhanced conceptual RobotStudio layout that provides a more detailed representation of the automation cell, including updated reach studies, device positioning, and process flow considerations. Any deviations from the original proposal, whether driven by clarified requirements, customer-provided details, or engineering constraints will be discussed and agreed upon during this phase. If there is significant scope change a COR will be required.

Typical Concept Engineering Deliverables:

- Conceptual System Layout (RobotStudio)
- Refined System Concept and Workflow Alignment
- Documented Open Items and Gap Closure
- Concept Engineering Review & Approval

5.5 Detailed Design

Technical specifications of the system are defined, and procurement will formally initiate. The functional design specification document will be generated and require approval by BASF. Timely approval is critical for the project timescale. It is anticipated that approval will take 1 week. The design is 'frozen' upon approval.

- Functional Design Specification
- Functional Design Specification Approval
- Final Design of the System

5.6 System Assembly

The system assembly function varies from project to project, but typically includes all or most of the following:

- Set-up of components.
- Mechanical, electrical, and pneumatic assembly.
- Functional Test and debug.
- Construction of mockups for demonstration (not included).
- Acceptance runoff (function Test).
- Tear down and packing for shipment.

5.7 Pre-FAT Debug and Validation

Prior to the Factory Acceptance Test (FAT), the system undergoes an internal debug and validation phase to verify that all hardware and software components operate correctly as an integrated system. During this stage, ABB engineers test robotic motion, device communication, automation sequences, safety functions, and the operator interface. Individual instruments and subsystems are verified for proper connectivity and control, and their interactions within the automation workflow are confirmed to ensure the system responds as intended.

This phase also focuses on identifying and resolving technical issues that could affect system operation prior to FAT. Engineers perform dry runs of automated sequences to verify device coordination, motion paths, timing parameters, and system logic. Adjustments may be made to optimize system performance and ensure stable and reliable operation. Completing these checks prior to FAT helps confirm that the system hardware, control architecture, and automation framework are functioning correctly and are ready for formal acceptance testing.

During the pre-FAT debug and validation phase, if either ABB or BASF identifies that specific needs cannot be met with the existing tools or processes, a formal change order process (COR) will be initiated. ABB project execution team will conduct a thorough assessment to understand the implications of the identified gaps, including potential impacts on project scope, timelines, and costs. Open communication will be maintained with the customer to ensure transparency regarding the needed adjustments. Any necessary changes will be documented through a change order, ensuring both parties agree on the revised requirements before proceeding with the FAT.

5.8 Factory Acceptance Test (FAT)

The Factory Acceptance Test (FAT) will be conducted at an ABB Robotics facility prior to shipment of the system. The purpose of FAT is to verify that the system hardware, automation controls, safety functions, and integrated devices operate according to the agreed system design and functional requirements. During this stage, ABB engineers will execute a series of predefined test procedures to demonstrate system operation, verify communication between integrated devices, and confirm proper execution of the automated workflows within the defined scope of the project.

ABB will prepare an FAT document that outlines the test procedures, acceptance criteria, and system functions to be verified during the runoff. The successful completion of these tests will constitute system runoff and acceptance at the ABB facility. FAT activities will be executed according to a schedule agreed upon by ABB and BASF. The following items describe important conditions and requirements related to preparation and execution of the FAT:

- Process function testing at the ABB facility is primarily limited to mechanical verification and water-based testing where applicable. When representative process verification is required, the system may be tested using water, safe substitute materials, or approved chemicals that reasonably simulate the intended application. Any chemicals used for testing must be supplied by the customer and must receive prior written approval from both the ABB Project Manager and ABB Environmental Health & Safety (EHS) before shipment to the ABB facility. These requirements are in place to ensure personnel safety, maintain compliance with applicable safety regulations, and support safe operation of the system during factory testing activities.
- Any discrepancies identified in documentation, test procedures, or system performance during testing will be reviewed and addressed through a defined correction plan. If such discrepancies are determined to result from non-ABB hardware, equipment, devices, or software selected or specified by Company Name, and additional engineering effort is required to achieve the functional requirements necessary to pass FAT, ABB reserves the right to submit a Change Order Request (COR) to cover the additional work required.
- FAT will be conducted using representative items over one 8-hour testing day. Additional testing or extended testing time may be requested by Company Name but must be communicated to ABB prior to design freeze and incorporated into the project scope through a COR. Any testing, schedule extensions, or scope changes beyond the agreed FAT protocol and timeline may require a separate change order and may incur additional costs. The specific testing activities outlined in the following table will be further defined and agreed upon during project execution.

Test Category	Description of FAT
Automation Process Testing	Verify system's ability to function as described in Section 1 following the automation process, hardware, and cycle time within the project plan timeframe for the FAT. As the processes described in these sections are extensive, the FAT will focus on representative, simplified tests that can be executed within the allocated timeframe. The results will be extrapolated to validate the overall cycle time.
Control System and PLC Logic Testing	Test control logic, alarms, interlocks, and emergency stop functionalities to meet operational sequences.
User Interface (HMI) Testing	Validate that HMI is intuitive, responsive, and displays real-time data and system statuses accurately.
Communication Protocols Testing	Test communication links between components (PLCs, sensors, actuators) for proper data transmission.
Electrical Systems Testing	Verify power supply, voltage levels, wiring, and signal integrity across all electrical components.
Mechanical Integrity Testing	Check mechanical components (actuators, conveyors) for correct installation, alignment, and operation.
Safety and Emergency Systems Testing	Test emergency stops, safety doors, and interlocks, ensuring system shutdown and restart safety.
Data Logging and Reporting	Test system's ability to log data and generate reports in the required format for analysis and review.
Power Loss and Recovery Testing	Simulate power failure and verify safe system recovery and restart

5.9 Site Acceptance Test (SAT)

The Site Acceptance Test (SAT) will be conducted at the BASF facility following installation of the system. The purpose of SAT is to verify that the system hardware, automation controls, safety functions, and integrated devices operate correctly within the customer's operational environment and according to the agreed system design and functional requirements. During this stage, ABB engineers will execute predefined test procedures to confirm proper system installation, verify communication with integrated equipment and facility infrastructure, and demonstrate that the automated workflows perform as intended under site operating conditions.

ABB will prepare an SAT document that outlines the test procedures, acceptance criteria, and system functions to be verified during the on-site testing. The successful completion of these tests will constitute system acceptance at the BASF facility. SAT activities will be executed according to a schedule agreed upon by ABB and BASF. The following items describe important conditions and requirements related to preparation and execution of the SAT:

- All materials required for SAT, including products, samples, chemicals, and consumables, must be supplied by the customer. Materials used during SAT should closely represent the products and process materials intended for normal system operation and should be similar to the materials used during Factory Acceptance Testing (FAT). All materials must be provided in formats compatible with the system hardware and consumables to ensure proper handling and operation during testing.
- ABB engineering support on-site for SAT is allocated for a limited period of 3 weeks as defined in the project scope. If the materials provided during SAT differ significantly from those used during factory testing or from those originally specified for the system—such as differences in density, viscosity, particle content, or other relevant properties—additional time may be required to evaluate system behavior and adjust operating parameters. In such cases, ABB may submit a Change Order Request (COR) to cover any additional engineering effort required.
- ABB is not responsible for any loss or consumption of customer products, samples, reagents, or consumables used during testing activities.
- Any discrepancies identified in documentation, test procedures, or system performance during SAT will be reviewed and addressed through a defined correction plan. If such discrepancies are determined to result from non-ABB hardware, equipment, devices, software, or facility conditions provided or selected by Company Name (for example facility readiness, IT infrastructure availability, or changes to specified consumables or materials), ABB reserves the right to submit a Change Order Request (COR) to cover the additional work required.
- Final acceptance and formal sign-off for SAT is expected to occur within one week from the start of SAT in order to avoid delays or extensions to the project schedule. SAT includes expanded testing compared to FAT to verify system functionality and performance under site conditions; however, all testing activities must be completed within the allocated timeframe. Any testing, schedule extensions, or scope changes



beyond the agreed SAT protocol and timeline may require a separate change order and may incur additional costs.

Test Category	Description of SAT
Installation Check	Verify proper installation of system components, excluding tools.
Power-Up Test	Ensure system powers up and initializes correctly based on start-up instruction or product manual and work instructions.
Automation Process Testing	Verify system's ability to function as described in Section 1 following the automation process, hardware, and cycle time within the project plan timeframe for the FAT. As the processes described in these sections are extensive, the FAT will focus on representative, simplified tests that can be executed within the allocated timeframe. The results will be extrapolated to validate the overall cycle time.
Control System and PLC Logic Testing	Test control logic, alarms, interlocks, and emergency stop functionalities to meet operational sequences.
User Interface (HMI) Testing	Validate that HMI is intuitive, responsive, and displays real-time data and system statuses accurately.
Communication Protocols Testing	Test communication links between components (PLCs, sensors, actuators) for proper data transmission.
Electrical Systems Testing	Verify power supply, voltage levels, wiring, and signal integrity across all electrical components.
Mechanical Integrity Testing	Check mechanical components (actuators, conveyors) for correct installation, alignment, and operation.
Safety and Emergency Systems Testing	Test emergency stops, safety doors, and interlocks, ensuring system shutdown and restart safety.
Data Logging and Reporting	Test system's ability to log data and generate reports in the required format for analysis and review.



Power Loss and Recovery Testing	Simulate power failure and verify safe system recovery and restart
Documentation Review	Review system manuals and records, excluding tool-specific documents.
Operator Training	Train operators on system use for sample handling, excluding tool training.

5.10 Request for Changes – CORs

Upon reception of any request for modification, ABB Project manager will acknowledge and give a date for the answer / quote. Only the modification request acknowledged by ABB project manager will be applied in the project. Changes to concept design after the Design Freeze Phase most likely will have significant changes to project cost and timeline.

A document would be formatted for information exchange between BASF and ABB with queries / clarifications and requests/approvals, it's aimed to clear issues occur possibly and to agree solutions before proceeding with actions related to changes of any kind.

5.11 Communication

Effective communication is essential to the successful execution of the project. ABB Robotics LLC will facilitate regular coordination meetings with BASF to ensure alignment, timely decision-making, and continuous progress throughout all phases of the project.

During the Concept Engineering and Detailed Design phases, weekly 1-hour project meetings will be held to review open items, track progress, and resolve any pending issues. As the project moves into Manufacturing and Commissioning, meeting frequency will transition to a bi-weekly cadence, with additional sessions scheduled as needed. For the final stages of the project, including preparation for FAT, SAT, and system handover, meetings will return to a weekly rhythm to ensure full alignment and readiness.

Additionally, the ABB project team may request supplementary technical meetings when needed to review specific aspects of the solution with the appropriate stakeholders from Company Name and any relevant third-party suppliers. These sessions ensure technical alignment, timely decisions, and clarity on interfaces or dependencies that may impact system performance or project schedule.

Meeting schedules and durations may be adjusted at the discretion of the ABB project team to support project needs, accommodate dependencies, or respond to changes in scope or timeline.

Project documentation and file sharing will be managed through a dedicated SharePoint folder or another mutually agreed method defined during the project kickoff. This shared environment will serve as the central repository for documents, drawings, schedules, and records to ensure version control and easy access for both teams.

5.12 Documentation

ABB Robotics LLC agrees to furnish mechanical layout drawings, all electrical schematics, I/O charts, and hardware schematic drawings in “.pdf” format.

As part of the project handover, ABB will supply a comprehensive user manual including:

- **System Overview & Architecture** — functional diagram, major components, and integrated subsystems (As a reference to functional specification document)

- **Operator Procedures** — step-by-step workflows for routine operation expected by the user, including configuration modes and parameter adjustment
- **Safety & Compliance** — safety system design, interlock logic, emergency procedures, and applicable safety certifications
- **Troubleshooting & Error Handling** — common error messages, diagnostic steps, and recovery procedures
- **Maintenance & Calibration** — preventive maintenance schedules, tool changeout procedures, sensor calibration, and spare parts requirements
- **HMI/Control Interface Guide** — navigation, status displays, and data export functionality (if applicable)
- **Integration Reference** — external system interfaces, data protocols, and third-party software integration points (if applicable)

Certain other proprietary design information will not be supplied. Such is the case with "off-the-shelf" products, which contain no custom design content. Wherever this occurs Customer will be furnished a complete bill of material and assembly drawing (in vendor provided format) describing all replaceable items.

5.13 Additional System Commissioning and Start-Up Support

(To be purchased separately see option)

It is assumed that BASF will purchase additional time as needed to support the startup and launch phases of this project if delays occur that are not the fault of ABB.

At additional cost ABB Robotics LLC can also supply one (1) field service technician, eight hours a day, during one (1) shift for a period of time that was outlined in the investment summary (not including weekends or holidays). This time is to be used for commissioning and start up support of the machine, system programming, start-up assistance, travel and for the verification and assistance of product capability. The utilization of the manpower will be distributed at the discretion of the ABB / BASF team.

Overtime and weekend work (if it is required or requested) will count as time and one half in terms using up the man-day allotment. At the completion of the allotted time, adjustments will be made for any time spent by ABB Robotics LLC technicians correcting equipment problems due to faulty design. Beyond that, additional commissioning time required must be purchased at ABB Robotics LLC

5.14 Clarifications and Assumptions

Equipment is delivered to site and installation of hardware commences. Prior to delivery, a site survey by ABB (virtually or in person) to ensure the site is ready for commencement of work. **ABB engineers are not responsible for any facility modifications prior to installing. Any site modification must be completed at BASF facility prior to ABB commissioning.**



ABB has the right to refuse to commission the system until the installation site is properly prepared and/or the environmental conditions are following ABB health and safety regulations.

1. ABB Robotics LLC process assumes zero interference with process equipment that is furnished by others.
2. ABB's scope of supply is limited to this proposal and cannot accept responsibility for the functionality of equipment furnished by others.
3. BASF is to provide labor and install all safety equipment not included in this proposal to meet specific plant safety specifications and OSHA regulations.
4. Our proposal is based on only the information we have received to date. Should new details surface after submittal of this proposal, or receipt of the order, adjustments in concepts and price may be necessary.
5. The authority to patent any special ABB Robotics LLC designed equipment must be negotiated at time of order placement. All patent rights are reserved by ABB Robotics LLC unless written approval is received from ABB prior to applying for any and all patents of ABB designed equipment.
6. It is ABB Robotics LLC understanding that (if necessary) the entire system will be floor mounted by BASF
7. No part tryout has been included in this proposal.
8. All project related meetings will be held at ABB Robotics LLC in Auburn Hills MI.

6 BASF Responsibilities:

It will be the responsibility of BASF to provide the following:

1. BASF is responsible for any additional safety equipment, safety equipment integration and documentation to the ABB furnished equipment to meet or exceed ANSI-RIA, OSHA, and/or other local equipment safety requirements including, but not limited to:
 - a. Additional Risk assessments.
 - b. Additional safety hardware.
 - c. Hardware integration.
 - d. Testing and verification of safety and cell operations.
 - e. Safety documents.
 - f. Lock-out tag out documents.
 - g. Suitable location of equipment, and or any other hazards associated with the equipment location.
2. BASF is responsible for sourcing all laboratory equipment and confirming that the equipment selected is automation friendly and meets the necessary process requirements.
3. BASF is responsible for depalletizing the equipment and moving it to the final installation location within their facility.
4. BASF is responsible for safety equipment not included in this proposal. There is no safety equipment integration included in this proposal.
5. BASF is responsible for all process development for parts and tooling. Specific process parameters are to be developed by Customer Name personnel.
6. BASF is responsible for ventilation of the cell.
7. Preparation of the plant floor and the surrounding area to accept the system. This includes the removal of all equipment which is not required for the operation of, or equipment interfering with the system.
8. Leveling and floor anchoring of ABB Robotics LLC provided machines and equipment and any reinforcing of the existing floor that is necessary.
9. The field wiring (material and labor) and location of any or all other machinery, equipment, conveyors, pallet stops, lift, and locates, signals, interfacing, design, re-design, build costs or any other expenses or liability incurred because of use or implementation of the aforementioned ABB Robotics LLC equipment, or ABB Robotics LLC provided services.



10. Provide time for operators and skilled trade's personnel to attend scheduled training classes on operation and maintenance of the equipment if purchased as an option.
11. Procurement of all consumable materials required for the operation of the system. The consumable material must be appropriate to what was scoped.
12. Procurement or supplier of all sample or product material that will be used on the system. Sample or product must be fresh and represent the material that are used in production.
13. Procurement or supplier of all reagents, standards, and other liquid material that will be used on the system. These materials must be fresh and in the correct container to represent the material that are used in production.
14. Disposal and shipping of any of the above materials and / or containers as required.
15. All necessary hanger support beams and special fabrications for wire ways.
16. Any field modifications to the system guarding not anticipated and performed while at ABB Robotics LLC facility.
17. All crating, shipping costs, freight duties, tariffs, country fees, taxes, or any other expenses attributed to delivery to BASF facilities.
18. All crating, duties, shipping costs, and taxes for BASF supplied materials and for the system.
19. Any other peripheral equipment, to complete the system, which is not specified in this proposal.
20. Training on the handling of hazardous materials not tested at the ABB facility, as well as the provision and safe use of customer-supplied equipment and PPE at the customer site, shall be the responsibility of the customer. ABB personnel will not conduct system testing involving hazardous substances or customer equipment unless such training and protective measures are confirmed to be in place.
21. Any safety equipment or person protective equipment (PPE) for the ABB employees or contractors that will be onsite for a visit or for installation, commissioning, SAT, UAT, or any other time ABB employees or contractors are onsite.

7 ABB ROBOTICS LLC – ROBOTICS BUSINESS AREA US GENERAL TERMS AND CONDITIONS OF SALE

1. General.

The terms and conditions contained herein, together with any additional or different terms contained in ABB's proposal, quotation and/or invoice ("Proposal"), if any, submitted to Purchaser (which Proposal, Policies, Addendum(s), if any, submitted to Purchaser shall control over any conflicting terms), constitute the entire agreement (the "Agreement") between the parties with respect to the purchase order and supersede all prior communications and agreements regarding the purchase order. Acceptance by ABB of the purchase order, or Purchaser's acceptance of ABB's Proposal, is expressly limited to and conditioned upon Purchaser's acceptance of these terms and conditions, payment for or acceptance of any performance by ABB being acceptance. These terms and conditions may not be changed or superseded by any different or additional terms and conditions proposed by Purchaser to which terms ABB hereby objects. Unless the context otherwise requires, the term "Equipment" as used herein means all of the equipment, parts, accessories sold, and all software and software documentation, if any, licensed to Purchaser by ABB ("Software") under the purchase order. Unless the context otherwise requires, the term "Services" as used herein means all labor, supervisory, technical and engineering, installation, repair, consulting or other services provided by ABB under the purchase order. As used herein, the term "Purchaser" shall also include the initial end user of the Equipment and/or services; provided, however, that Article 14(a) shall apply exclusively to the initial end user.

2. Prices.

(a) Unless otherwise specified in writing, all Proposals expire thirty (30) days from the date thereof and may be modified or withdrawn by ABB before receipt of Purchaser's conforming acceptance. All quoted prices are subject to revision at any time in the event of any increase in raw material, energy costs or governmental actions such as tariffs.

(b) Unless otherwise stated herein, Services prices are based on normal business hours (8 a.m. to 5 p.m. Monday through Friday). Overtime and Saturday hours will be billed at one and one-half (1 1/2) times the hourly rate; and Sunday hours will be billed at two (2) times the hourly rate; holiday hours

will be billed at three (3) times the hourly rate. If a Services rate sheet is attached hereto, the applicable Services rates shall be those set forth in the rate sheet. Rates are subject to change without notice.

(c) The price does not include any federal, state or local property, license, privilege, sales, use, excise, gross receipts, or other like taxes which may now or hereafter be applicable. Purchaser agrees to pay or reimburse any such taxes which ABB or its suppliers are required to pay or collect. If Purchaser is exempt from the payment of any tax or holds a direct payment permit, Purchaser shall, upon purchase order placement, provide ABB a copy, acceptable to the relevant governmental authorities of any such certificate or permit.

(d) The price includes customs duties and other importation or exportation fees, if any, at the rates in effect on the date of ABB's Proposal. Any change after that date in such duties, fees, or rates, shall increase the price by ABB's additional cost.

(e) Overtime, exchange rates fluctuate and may impair a party's ability to operate under the terms of the Agreement. In such event the parties will meet to determine if there is a need for an equitable adjustment.

3. Payment.

(a) Unless specified to the contrary in writing by ABB, payment terms are net cash, payable without offset, in United States Dollars, 30 days from date of invoice by wire transfer to the account designated by ABB in the Proposal. ABB is not required to commence or continue its performance unless and until invoiced payments have been received in a timely fashion. For each day of delay in receiving required payments, ABB shall be entitled to a matching extension of the schedule.

(b) If in the judgment of ABB, the financial condition of Purchaser at any time prior to delivery does not justify the terms of payment specified, ABB may require payment in advance, payment security satisfactory to ABB and suspend its performance until said advance payment or payment security is received or may terminate the purchase order, whereupon ABB shall be entitled to receive reasonable

cancellation charges. If delivery is delayed by Purchaser, payment shall be due on the date ABB is prepared to make delivery. Delays in delivery or non-conformities in any installments delivered shall not relieve Purchaser of its obligation to accept and pay for remaining installments.

(c) Purchaser shall pay, in addition to the overdue payment, a late charge equal to the lesser of 1 1/2% per month or any part thereof or the highest applicable rate allowed by law on all such overdue amounts plus ABB's attorneys' fees and court costs incurred in connection with collection. If Purchaser fails to make payment of any amounts due under any purchase order and fails to cure such default within ten (10) days after receiving written notice specifying such default, then ABB may by written notice, at its option, suspend its performance under the purchase order until such time as the full balance is paid or terminate the purchase order, as of a date specified in such notice. In the event of suspension, cancellation or termination hereunder, ABB will be entitled to recover all costs for work performed to date, costs associated with suspension, cancellation or termination of the work and all other costs recoverable at law.

4. Changes.

(a) Any changes requested by Purchaser affecting the ordered scope of work must first be reviewed by ABB and any resulting adjustments to affected provisions, including price, schedule, and guarantees mutually agreed in writing prior to implementation of the change.

(b) ABB may, at its expense, make such changes in the Equipment or Services as it deems necessary, in its sole discretion, to conform the Equipment or Services to the applicable specifications. If Purchaser objects to any such changes, ABB shall be relieved of its obligation to conform to the applicable specifications to the extent that conformance may be affected by such objection.

5. Delivery.

- (a) All Equipment manufactured, assembled or warehoused in the continental United States is delivered FCA ABB point of shipment, Incoterms® 2020 unless otherwise mutually agreed in writing. Equipment shipped outside the continental United States is delivered FCA Incoterms® 2020 United States port of export unless otherwise mutually

agreed in writing. Purchaser shall be responsible for any and all demurrage or detention charges.

- (b) If the scheduled delivery of Equipment is delayed by Purchaser or by Force Majeure, ABB may move the Equipment to storage for the account of and at the risk of Purchaser whereupon it shall be deemed to be delivered.
- (c) Shipping and delivery dates are contingent upon Purchaser's timely approvals and delivery by Purchaser of any documentation required for ABB's performance hereunder.
- (d) Claims for shortages or other errors in delivery must be made in writing to ABB within ten (10) days of delivery. Equipment may not be returned except with the prior written consent of and subject to terms specified by ABB. Claims for damage after delivery shall be made directly by Purchaser with the common carrier.

6. Title & Risk of Loss.

Except with respect to Software (for which title shall not pass, use being licensed) title to Equipment shall transfer to Purchaser upon delivery according to the applicable freight term. Notwithstanding any agreement with respect to delivery terms or payment of transportation charges, risk of loss or damage with respect to the sale of Equipment shall pass from ABB to Purchaser at delivery as defined in this Agreement.

7. Inspection, Testing and Acceptance.

(a) Any inspection by Purchaser of Equipment on ABB's premises shall be scheduled in advance to be performed during normal working hours and subject to rules and regulations in place at the ABB premises.

(b) If the purchase order provides for factory acceptance testing, ABB shall notify Purchaser when ABB will conduct such testing prior to shipment. Unless Purchaser states specific objections in writing within ten (10) days after completion of factory acceptance testing, completion of the acceptance test constitutes Purchaser's factory acceptance of the Equipment and its authorization for shipment.

(c) If the purchase order provides for site acceptance testing, testing will be performed by ABB personnel to verify that the Equipment has arrived at site complete, without physical damage, and in good operating condition. Completion of site acceptance testing constitutes full and final acceptance of the Equipment. If, through no fault of

ABB, acceptance testing is not completed within thirty (30) days after arrival of the Equipment at the site, the site acceptance test shall be deemed completed and the Equipment shall be deemed accepted.

8. Warranties and Remedies.

(a) Equipment and Services Warranty. ABB warrants that Equipment (excluding Software, which is warranted as specified in paragraph (d) below) shall be delivered free of defects in material and workmanship and that Services shall be free of defects in workmanship. The Warranty Remedy Period for Equipment (excluding Software and Spare Parts) shall end twelve (12) months after installation or eighteen (18) months after date of shipment, whichever first occurs. The Warranty Remedy Period for new spare parts shall end twelve (12) months after date of shipment. If the purchase order includes the sale of refurbished or repaired parts, the Warranty Remedy Period for such parts shall end ninety (90) days after date of shipment. The Warranty Remedy Period for Services shall end ninety (90) days after the date of completion of Services.

(b) Equipment and Services Remedy. If a nonconformity to the foregoing warranty is discovered in the Equipment or Services during the applicable Warranty Remedy Period, as specified above, under normal and proper use and provided the Equipment has been properly stored, installed, operated and maintained and written notice of such nonconformity is provided to ABB promptly after such discovery and within the applicable Warranty Remedy Period, ABB shall, at its option, either (i) repair or replace the nonconforming portion of the Equipment or re-perform the nonconforming Services or (ii) refund the portion of the price applicable to the nonconforming portion of Equipment or Services. If any portion of the Equipment or Services so repaired, replaced or re-performed fails to conform to the foregoing warranty, and written notice of such nonconformity is provided to ABB promptly after discovery and within the original Warranty Remedy Period applicable to such Equipment or Services or thirty (30) days from completion of such repair, replacement or re-performance, whichever is later, ABB will repair or replace such nonconforming Equipment or re-perform the nonconforming

Services. The original Warranty Remedy Period shall not otherwise be extended.

(c) Exceptions. ABB shall not be responsible for providing temporary power, removal, installation, reimbursement for labor costs or working access to the nonconforming Equipment, including disassembly and re-assembly of non-ABB supplied equipment, or for providing transportation to or from any repair facility, or for any other expenses incurred in connection with the repair or replacement, all of which shall be at Purchaser's risk and expense. ABB shall have no obligation hereunder with respect to any Equipment which (i) has been improperly repaired or altered; (ii) has been subjected to misuse, negligence or accident; (iii) has been used in a manner contrary to ABB's instructions; (iv) is comprised of materials provided by or a design specified by Purchaser; or (v) has failed as a result of ordinary wear and tear. Equipment supplied by ABB but manufactured by others is warranted only to the extent of the manufacturer's warranty, and only the remedies, if any, provided by the manufacturer will be allowed.

(d) Software Warranty and Remedies. ABB warrants that, except as specified below, the Software will, when properly installed, execute in accordance with ABB's published specification. If a nonconformity to the foregoing warranty is discovered during the period ending one (1) year after the date of shipment and written notice of such nonconformity is provided to ABB promptly after such discovery and within that period, including a description of the nonconformity and complete information about the manner of its discovery, ABB shall correct the nonconformity by, at its option, either (i) modifying or making available to the Purchaser instructions for modifying the Software; or (ii) making available at ABB's facility necessary corrected or replacement programs. ABB shall have no obligation with respect to any nonconformities resulting from (i) unauthorized modification of the Software or (ii) Purchaser-supplied software or interfacing. ABB does not warrant that the functions contained in the software will operate in combinations which may be selected for use by the Purchaser, or that the software products are free from errors in the nature of what is commonly categorized by the computer industry as "bugs".

(e) THE FOREGOING WARRANTIES ARE EXCLUSIVE AND IN LIEU OF ALL OTHER WARRANTIES OF QUALITY AND PERFORMANCE, WHETHER WRITTEN, ORAL OR IMPLIED, AND ALL OTHER WARRANTIES INCLUDING ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USAGE OF TRADE ARE HEREBY DISCLAIMED. THE REMEDIES STATED HEREIN CONSTITUTE PURCHASER'S EXCLUSIVE REMEDIES AND ABB'S ENTIRE LIABILITY FOR ANY BREACH OF WARRANTY.

9. Intellectual Property Indemnification.

(a) ABB shall defend at its own expense any action brought against Purchaser alleging that the Equipment or the use of the Equipment to practice any process for which such Equipment is specified by ABB (a "Process") directly infringes a patent in effect in the United States, an European Union member state or the country of the Site (provided there is a corresponding patent issued by the U.S., UK or an EU member state), or any copyright or trademark registered in the country of the Site and to pay all damages and costs finally awarded in any such action, provided that Purchaser has given ABB prompt written notice of such action, all necessary assistance in the defense thereof and the right to control all aspects of the defense thereof including the right to settle or otherwise terminate such action on behalf of Purchaser.

(b) ABB shall have no obligation hereunder and this provision shall not apply to: (i) any other equipment or processes, including Equipment or Processes which have been modified or combined with other equipment or process not supplied by ABB; (ii) any Equipment or Process supplied according to a design, other than an ABB design, required by Purchaser; (iii) any products manufactured by the Equipment or Process; (iv) any use of the Equipment or Process contrary to ABB instructions; (v) any patent issued after the date hereof; or (vi) any action settled or otherwise terminated without the prior written consent of ABB.

(c) If, in any such action, the Equipment is held to constitute an infringement, or the practice of any Process using the Equipment is finally enjoined, ABB shall, at its option and its own expense, procure for Purchaser the right to continue using said Equipment; or modify or replace it with

non-infringing equipment or, with Purchaser's assistance, modify the Process so that it becomes non-infringing; or remove it and refund the portion of the price allocable to the infringing Equipment. THE FOREGOING PARAGRAPHS STATE THE EXCLUSIVE LIABILITY OF ABB AND EQUIPMENT MANUFACTURER FOR ANY INTELLECTUAL PROPERTY INFRINGEMENT.

(d) To the extent that said Equipment or any part thereof is modified by Purchaser, or combined by Purchaser with equipment or processes not furnished hereunder (except to the extent that ABB is a contributory infringer) or said Equipment or any part thereof is used by Purchaser to perform a process not furnished hereunder by ABB or to produce an article, and by reason of said modification, combination, performance or production, an action is brought against ABB, Purchaser shall defend and indemnify ABB in the same manner and to the same extent that ABB would be obligated to indemnify Purchaser under this "Intellectual Property Indemnification" provision.

10. Waiver of Consequential Damages.

In no event shall ABB, its suppliers or subcontractors be liable for special, indirect, incidental or consequential damages, whether in contract, warranty, tort, negligence, strict liability or otherwise, including, but not limited to, loss of profits or revenue, loss of data, loss of use, loss of use of any of the Equipment or any associated equipment, cost of capital, cost of substitute equipment, facilities or services, downtime costs, delays, and claims of customers of the Purchaser or other third parties for any damages.

11. Limitation of Liability.

(a) ABB's aggregate liability for all claims whether in contract, warranty, tort, negligence, strict liability, or otherwise for any loss or damage arising out of, connected with, or resulting from this Agreement or the performance or breach thereof, or from the design, manufacture, sale, delivery, resale, repair, replacement, installation, technical direction of installation, inspection, operation or use of any equipment covered by or furnished under this Agreement, or from any services rendered in connection therewith, shall in no case (except as

provided in the section entitled "Intellectual Property Indemnification") exceed the purchase order price.

(b) All causes of action against ABB arising out of or relating to this Agreement or the performance or breach hereof shall expire unless brought within one (1) year of the time of accrual thereof.

(c) In no event, regardless of cause, shall ABB be liable for penalties or penalty clauses of any description or for indemnification of Purchaser or others for costs, damages, or expenses arising out of or related to the Equipment and/Services.

12. Laws and Regulations.

ABB does not assume any responsibility for compliance with federal, state or local laws and regulations, except as expressly set forth herein, and compliance with any laws and regulations relating to the operation or use of the Equipment or Software is the sole responsibility of the Purchaser. All laws and regulations referenced herein shall be those in effect as of the Proposal date. In the event of any subsequent revisions or changes thereto, ABB assumes no responsibility for compliance therewith. If Purchaser desires a modification as a result of any such change or revision, it shall be treated as a change per Article 4. Nothing contained herein shall be construed as imposing responsibility or liability upon ABB for obtaining any permits, licenses or approvals from any agency required in connection with the supply, erection or operation of the Equipment. This Agreement shall in all respects be governed by, and construed, interpreted and enforced in accordance with the laws of the State of New York, USA, excluding its conflicts of laws rules and the provisions of the United Nations Convention on Contracts for the International Sale of Goods, and both parties hereby agree that any litigation concerning, arising out of, or related to this Agreement, whether claims are based on contract, tort, equity or otherwise, shall be conducted only in the state or federal courts functioning in the State of New York and waive the defense of an inconvenient forum in respect to any such litigation. If any provision hereof, partly or completely, shall be held invalid or unenforceable, such invalidity or unenforceability shall not affect any other provision or portion hereof and these terms shall be construed as if such

invalid or unenforceable provision or portion thereof had never existed.

13. OSHA.

ABB warrants that the Equipment will comply with the relevant standards of the Occupational Safety and Health Act of 1970 ("OSHA") and the regulations promulgated thereunder as of the date of the Proposal. Upon prompt written notice from the Purchaser of a breach of this warranty, ABB will replace the affected part or modify it so that it conforms to such standard or regulation. ABB's obligation shall be limited to such replacement or modification. In no event shall ABB be responsible for liability arising out of the violation of any OSHA standards relating to or caused by Purchaser's design, location, operation, or maintenance of the Equipment, its use in association with other equipment of Purchaser, or the alteration of the Equipment by any party other than ABB.

14. Software License.

(a) ABB owns all rights in or has the right to sublicense all of the Software, if any, to be delivered to Purchaser under this Agreement. As part of the sale made hereunder Purchaser hereby obtains a limited license to use the Software, subject to the following: (i) the Software may be used only in conjunction with equipment specified by ABB; (ii) the Software shall be kept strictly confidential; (iii) the Software shall not be copied, reverse engineered, or modified; (iv) the Purchaser's right to use the Software shall terminate immediately when the specified equipment is no longer used by the Purchaser or when otherwise terminated, e.g. for breach, hereunder; and (v) the rights to use the Software are non-exclusive and non-transferable, except with ABB's prior written consent.

(b) Nothing in this Agreement shall be deemed to convey to Purchaser any title to or ownership in the Software or the intellectual property contained therein in whole or in part, nor to designate the Software a "work made for hire" under the Copyright Act, nor to confer upon any person who is not a named party to this Agreement any right or remedy under or by reason of this Agreement. In the event of termination of this License, Purchaser shall immediately cease using the Software and, without retaining any copies, notes or excerpts thereof,

return to ABB the Software and all copies thereof and shall remove all machine-readable Software from all of Purchaser's storage media.

15. Intellectual Property, Inventions and Information.

- (a) "Intellectual Property Rights" means all current and future rights in copyrights, trade secrets, trademarks, mask works, patents, design rights, trade dress, and any other intellectual property rights that may exist anywhere in the world, including, in each case whether unregistered, registered or comprising an application for registration, and all rights and forms of protection of a similar nature or having equivalent or similar effect to any of the foregoing. "Technology" means all inventions, discoveries, ideas, concepts, methods, code, executables, manufacturing processes, unique compositions, mask works, designs, marks, and works of authorship fixed in the medium of expression, and materials pertaining to any of the preceding; whether or not patentable, copyrightable or subject to other forms of protection.
- (b) ABB shall maintain all right, title and interest in any Technology and Intellectual Property Rights that ABB owned, created, conceived or discovered prior to entering into this Agreement, or owns, creates or discovers separately from the activities contemplated by this Agreement. Unless otherwise agreed in writing by ABB and Purchaser, ABB shall have all right, title and interest in any Technology and Intellectual Property Rights that ABB creates, conceives or discovers in furtherance of this Agreement, and ABB shall have all right, title and interest in any Technology and Intellectual Property Rights embodied in the Equipment and Services. Any design, manufacturing drawings or other information submitted to the Purchaser remains the exclusive property of ABB. Purchaser shall not, without ABB's prior written consent, copy or disclose such information to a third party, unless required by a public information request from a governmental body. Such information shall be used solely for the operation or maintenance of the Equipment and not for any other purpose, including the duplication thereof in whole or in part.

16. Force Majeure.

ABB shall neither be liable for loss, damage, detention or delay nor be deemed to be in default for failure to perform when prevented from doing so by causes beyond its reasonable control including but not limited to acts of war (declared or undeclared), delays attributable to outbreaks, epidemics and pandemics, Acts of God, fire, strike, labor difficulties, acts or omissions of any governmental

authority or of Purchaser, compliance with government regulations, insurrection or riot, embargo, delays or shortages in transportation or inability to obtain necessary labor, materials, or manufacturing facilities from usual sources or from defects or delays in the performance of its suppliers or subcontractors due to any of the foregoing enumerated causes. In the event of delay due to any such cause, the date of delivery will be extended by period equal to the delay plus a reasonable time to resume production, and the price will be adjusted to compensate ABB for such delay.

17. Cancellation.

Special order, custom designed, and made-to-order Equipment are non-cancelable and non-returnable. Any other purchase order may be cancelled by Purchaser only upon prior written notice and payment of termination charges as set forth in the cancellation schedule included in the Proposal or payment of, including but not limited to, the purchase price of the work performed prior to the effective date of notice of termination, the costs identified to the purchase order incurred by ABB for work not completed, and all expenses incurred by ABB attributable to the termination, plus a fixed sum of ten (10) percent of the final total price to compensate for disruption in scheduling, planned production and other indirect costs.

18. Termination.

- (a) No termination by Purchaser for material default shall be effective unless, within fifteen (15) days after receipt by ABB of Purchaser's written notice specifying such default, ABB shall have failed to initiate and pursue with due diligence correction of such specified default.
- (b) If the event of termination for a material default, ABB shall reimburse Purchaser the difference between that portion of the Agreement price allocable to the terminated scope and the actual amounts reasonably incurred by Purchaser to complete that scope, and Purchaser shall pay to ABB the portion of the Agreement price allocable to Equipment completed and any amounts due for Services performed before the effective date of termination.
- (c) ABB may terminate the Agreement (or any affected portion thereof) immediately for cause if Purchaser becomes insolvent/bankrupt, or materially breaches the Agreement, including, but not limited to, failure or delay in Purchaser making any payment when due, or fulfilling any payment conditions.

19. Export & Trade Control.

(a) Sanctions and trade control laws and regulations: any applicable laws, regulations, or administrative or regulatory decisions or guidelines that sanction, prohibit or restrict

certain activities including, but not limited to, (i) import, export, re-export, transfer, or trans-shipment of goods, services, technology, or software; (ii) financing of, investment in, or direct or indirect transactions or dealings with certain countries, territories, regions, governments, projects, or specifically designated persons or entities, including any future amendments to these provisions; or (iii) any other laws, regulations, administrative or regulatory decisions, or guidelines adopted, maintained, or enforced by any Sanctions Agency on or after the date of this Agreement (collectively, "Trade Control Laws").

(b) Restricted Person means any entity or person included on a list (including U.S. and EU lists) of targeted parties, blocked parties, or persons subject to asset-freezing or other restrictions introduced under any applicable Trade Control Laws (and includes any entity that is directly or indirectly owned fifty (50) percent or more, in the aggregate or individually, or otherwise controlled by any Restricted Person).

(c) Each party represents and warrants that, to the best of its knowledge, at the date of this Agreement neither it, nor any of their respective directors or officers are a Restricted Party. Each party agrees that it shall promptly notify the other party if it becomes a Restricted Party.

(d) If, as a result of (i) Trade Control Laws issued or amended after the date of this Agreement, (ii) Purchaser becoming a Restricted Party, or (iii) any necessary export license or authorization from a Sanctions Agency is not granted, the performance by ABB or by any affiliates or third parties engaged in any manner in relation to the Agreement becomes illegal or impracticable, ABB shall, as soon as reasonably practicable, give written notice to Purchaser of its inability to perform or fulfill such obligations. Once such notice has been received by Purchaser, ABB shall be entitled to either immediately suspend the performance of the affected obligation under the Agreement until such time as ABB may lawfully discharge such obligation or unilaterally terminate the Agreement in whole or in

part from the date specified in the said written notice or from any subsequent date thereafter. ABB will not be liable to Purchaser for any costs, expenses or damages associated with such suspension or termination of the Agreement.

(e) In the event of suspension or termination as set out in Section 19(d) above, ABB shall be entitled to payment as set out in this Agreement and any reasonable associated costs necessarily incurred by ABB in regard to such suspension or termination including, but not limited to, all reasonable costs associated with suspending or terminating any sub-contract placed or committed for goods or services in connection with this Agreement.

(f) Purchaser represents and warrants that the Equipment and Services provided hereunder, and the "direct product" thereof are intended for civil use only and will not be used, directly or indirectly, for the production of chemical or biological weapons or of precursor chemicals for such weapons, or for any direct or indirect nuclear end use. Purchaser agrees not to disclose, use, export or re-export, directly or indirectly, any information provided by ABB or the "direct product" thereof as defined in the Export Control Regulations of the United States Department of Commerce, except in compliance with such Regulations. The Purchaser further represents that it will not directly or indirectly sell, export, re-export, release, transmit or otherwise transfer any items received from ABB to any Restricted Parties, or parties that operate, or whose end use will be, in a jurisdiction/region prohibited by ABB including Belarus, Crimea, Cuba, Iran, North Korea, Russia, Syria, as well as the Donetsk, Luhansk, Kherson, and Zaporizhzhia regions of Ukraine (such list may be amended by ABB at any time).

(g) Purchaser represents that it will not directly or indirectly sell, export, re-export, release, transmit or otherwise transfer any items received from ABB to any third party or country in violation of Trade Control Laws.

(h) If applicable, ABB shall file for a U.S. export license, but only after appropriate documentation for the license application has been provided by Purchaser. Purchaser shall furnish such documentation within a reasonable time after purchase order acceptance. Any delay in obtaining such license shall suspend performance of this Agreement by ABB. If an export license is not granted or, if once

granted, is thereafter revoked or modified by the appropriate authorities, this Agreement may be cancelled by ABB without liability for damages of any kind resulting from such cancellation. At ABB's request, Purchaser shall provide to ABB a Letter of Assurance and End-User Statement in a form reasonably satisfactory to ABB.

(i) If the Purchaser infringes any obligations in this Export and Trade Control clause in connection with the Agreement, the Purchaser must immediately notify ABB. Failure to comply with these trade compliance obligations shall be considered a material breach, and ABB shall have the right to unilaterally terminate the Agreement with immediate effect. Such termination would be without prejudice to all rights of recourse which could be exercised by ABB, and ABB shall not be liable to Purchaser for any claim, losses or damages whatsoever related to its decision to terminate performance under this provision. Further, Purchaser shall indemnify ABB for all liabilities, damages, costs, or expenses incurred as a result of any such violation, breach and/or termination of the Agreement. ABB may report such violations to relevant authorities as required by applicable Trade Control Laws.

(j) For the avoidance of doubt, no provision in this Agreement shall be interpreted or applied in a way that would require any party to do, or refrain from doing, any act which would constitute a violation of, or result in a loss of economic benefit under, applicable Trade Control Laws.

20. Bribery and Corruption & Human Rights

(a) Purchaser hereby warrants that it will not, directly or indirectly, and it has no knowledge that other persons will, directly or indirectly, make any payment, gift or other commitment to its customers, to government officials or to agents, directors and employees of ABB or any other party in a manner contrary to applicable laws (including but not limited to the Corruption of Foreign Public Officials Act (Canada), the Foreign Corrupt Practices Act (United States) and, where applicable, legislation enacted by member States and signatories implementing the OECD Convention on Combating Bribery of Foreign Officials) and shall comply with all relevant laws, regulations, ordinances and rules regarding bribery and corruption, anti-money laundering and anti-tax evasion.

(b) Nothing hereunder shall render ABB liable to reimburse Purchaser for any such consideration given or promised.

(c) Purchaser's material violation of any of the obligations contained in this Section 20 may be considered by ABB to be a material breach hereunder and shall entitle ABB to terminate this Agreement with immediate effect and without prejudice to any further right or remedies on the part of ABB hereunder or applicable law. Purchaser shall indemnify ABB for all liabilities, damages, costs or expenses incurred as a result of any such violation of the above-mentioned obligations and termination of this Agreement.

(d) Purchaser understands that ABB's Code of Conduct is available for consultation online at <http://www.abb.com/integrity>. Purchaser agrees to perform its contractual obligations hereunder with substantially similar standards of ethical behavior as those found in ABB's Code of Conduct.

(e) ABB has established the following reporting channels where Purchaser and its employees may report suspected violations of applicable laws, policies or standards of conduct:

Web portal: www.abb.com/integrity

Telephone: number specified on the above Web portal

Mail: address specified on the above Web portal

(f) Human rights and anti-modern slavery laws: including The Universal Declaration of Human Rights, the UN Guiding Principles on Business and Human Rights, the OECD Guidelines for Multinational Enterprises, the ILO Core Conventions on Labor Standards, the UK Modern Slavery Act and other similar human rights, anti-human trafficking and anti-modern slavery laws and regulations (collectively, "Human Rights Laws"). Both parties will comply with Human Rights Laws in connection with this Agreement. Both parties shall ensure that their respective employees, officers, directors, and any affiliates or third parties engaged in any manner in relation to this Agreement shall undertake to comply with all Human Rights Laws. Both parties confirm that they have not violated, shall not violate, and shall not cause the other party to violate any applicable Human Rights Laws in connection with this Agreement.

21. Assignment.

Any assignment of this Agreement or of any rights or obligations under the Agreement without prior written consent of ABB shall be void.

22. Nuclear.

Equipment and Services sold hereunder are not intended for use in connection with any nuclear facility or activity. Purchaser warrants that such Equipment and Services shall not be installed, used or applied in or in connection with (i) the design, production, use or storage of chemical, biological or nuclear weapons or their delivery systems, (ii) any military applications or (iii) the operation of any nuclear facilities including, but not limited to, nuclear power plants, nuclear fuel manufacturing plants, uranium enrichment plants, spent nuclear fuel stores and research reactors, and Purchaser shall not permit others to use Equipment or Services for such purposes, without the advance written consent of ABB. If, in breach of this, any such use occurs, ABB (and its parent, affiliates, suppliers and subcontractors) disclaims all liability for any nuclear or other damage, injury or contamination, including without limitation any physical damage to a nuclear facility itself, resulting from a nuclear incident and, in addition to any other rights of ABB, Purchaser shall indemnify and hold ABB (and its parent, affiliates, suppliers and subcontractors) harmless against all such liability including, but not limited to, any physical damage to the nuclear facility or surrounding properties, if any. Consent of ABB to any such use, if any, will be conditioned upon additional terms and conditions that ABB determines to be acceptable for protection against nuclear liability including but not limited to the requirement that the Purchaser and/or its end user customer shall have complete insurance protection against liability and property damage including without limitation physical damage to a nuclear facility itself or any surrounding properties, if any, resulting from a nuclear incident and shall indemnify ABB, its subcontractors, suppliers and vendors against all claims resulting from a nuclear incident including, but not limited to, any physical damage to the nuclear facility.

23. Resale.

If Purchaser resells any of the Equipment or Services, the sale terms shall limit ABB's liability to the buyer to the same extent that ABB's liability to Purchaser is limited hereunder. Additionally, if the end-user intends to use the Equipment or Services in connection with any nuclear facility or activity, the Purchaser shall require the end-user comply with the financial requirements under Price-Anderson Act (PAA) and secure a written release of liability which flows from the end-user to the benefit of ABB.

24. Environmental, Health and Safety Matters.

- (a) Purchaser shall be obligated to maintain safe working conditions at its facility or location (the "Site"), including the implementing of appropriate procedures regarding Hazardous Materials, confined space entry, and energization and de-energization of power systems (electrical, mechanical and hydraulic) using safe and effective lock-out/tag-out ("LOTO") procedures including physical LOTO or a mutually agreed upon alternative method.
- (b) Purchaser shall immediately advise ABB in writing of all applicable Site-specific health, safety, security and environmental requirements and procedures. Without limiting Purchaser's responsibilities hereunder, ABB has the right but not the obligation to, from time to time, review, audit and inspect applicable health, safety, security and environmental documentation, procedures and conditions at the Site.
- (c) If, in ABB's reasonable opinion, the health, safety, or security of personnel or the Site is, or is likely to be, imperiled by security risks, the presence of or threat of exposure to Hazardous Materials, or unsafe working conditions, ABB may, in addition to other rights or remedies available to it, remove some or all of its personnel from Site, suspend performance of all or any part of the purchase order, and/or remotely perform or supervise work. Any such occurrence shall be considered a Force Majeure event. Purchaser shall reasonably assist in ensuring the safe departure of personnel from the Site.
- (d) Purchaser shall not require or permit ABB's personnel to operate Purchaser's equipment at Site.
- (e) Purchaser will make its Site medical facilities and resources reasonably available to ABB personnel who need medical attention.
- (f) ABB has no responsibility or liability for the pre-existing condition of Purchaser's equipment or the Site, which is the sole responsibility of Purchaser. Prior to ABB starting any work at Site, Purchaser will provide documentation that identifies the presence and condition of any Hazardous Materials

existing in or about Purchaser's equipment or the Site that ABB may encounter while performing under this Agreement. The provision of such documentation shall in no way release Purchaser from its responsibility for said conditions. Purchaser shall disclose to ABB industrial hygiene and environmental monitoring data regarding conditions that may affect ABB's work or personnel at the Site. Purchaser shall keep ABB informed of changes in any such conditions.

- (g) ABB shall promptly notify Purchaser if ABB becomes aware of: (i) conditions at the Site differing materially from those disclosed by Purchaser, or (ii) previously unknown physical conditions at Site differing materially from those ordinarily encountered and generally recognized as inherent in work of the character provided for in the Agreement. If any such conditions cause an increase in ABB's cost of, or the time required for, performance of any part of the work under the Agreement, an equitable adjustment in price and schedule shall be made.
- (h) If ABB encounters Hazardous Materials in Purchaser's equipment or at the Site that require special handling or disposal, ABB is not obligated to continue work affected by the hazardous conditions. In such an event, Purchaser shall at its sole cost and expense eliminate the hazardous conditions in accordance with applicable laws and regulations so that ABB's work under the Agreement may safely proceed, and ABB shall be entitled to an equitable adjustment of the price and schedule to compensate for any increase in ABB's cost of, or time required for, performance of any part of the work. Purchaser shall properly store, transport and dispose of all Hazardous Materials introduced, produced or generated in the course of ABB's work at the Site.
- (i) Purchaser shall indemnify ABB for any and all claims, damages, losses, and expenses arising out of or relating to any Hazardous Materials which are or were (i) present in or about Purchaser's equipment or the Site prior to the commencement of ABB's work, (ii) improperly handled or disposed of by Purchaser or Purchaser's employees, agents, contractors or subcontractors, or (iii) brought, generated, produced or released on Site by parties other than ABB.

25. Confidentiality.

(a) ABB and Purchaser (as to information disclosed, the "Disclosing Party") may each provide the other party (as to information received, the "Receiving Party") with Confidential Information in connection with this Agreement. "Confidential Information" means (a) information that is designated in writing as "confidential" or "proprietary" by Disclosing

Party at the time of written disclosure, and (b) information that is orally designated as "confidential" or "proprietary" by Disclosing Party at the time of oral or visual disclosure and is confirmed to be "confidential" or "proprietary" in writing within fifteen (15) days after the oral or visual disclosure or (c) information by its nature or the circumstances surrounding disclosure is or reasonably should be understood to be confidential or proprietary to the Disclosing Party or its Affiliates. In addition, prices for Equipment and Services shall be considered ABB's Confidential Information.

(b) Receiving Party agrees: (i) to use the Confidential Information only in connection with the Agreement and use of Equipment and Services, (ii) to take reasonable measures to prevent disclosure of the Confidential Information to third parties, and (iii) not to disclose the Confidential Information to a competitor of Disclosing Party. Notwithstanding these restrictions, each party shall permit access to the other's Confidential Information only to its employees who: (i) reasonably require access to Confidential Information for purposes approved by this Agreement, and (ii) have undertaken a binding obligation of confidentiality with respect to the confidential information of others entrusted to him or her, and (iii) have been apprised of the confidentiality obligations hereunder. ABB may disclose Confidential Information to its affiliates and subcontractors in connection with performance of the purchase order. A Receiving Party may only disclose Confidential Information to any other third party with the prior written permission of Disclosing Party, and in each case, only so long as the Receiving Party obtains a non-disclosure commitment from any such third party that prohibits disclosure of the Confidential Information and provided further that the Receiving Party remains responsible for any unauthorized use or disclosure of the Confidential Information. Receiving Party shall upon request return to Disclosing Party or destroy all copies of Confidential Information except to the extent that a specific provision of the Agreement entitles Receiving Party to retain an item of Confidential Information. ABB may also retain one archive copy of Purchaser's Confidential Information.

(c) The obligations under this Article 25 shall not apply to any portion of the Confidential Information that: (i) is or becomes generally available to the

public other than as a result of disclosure by Receiving Party, its representatives or its affiliates; (ii) is or becomes available to Receiving Party on a non-confidential basis from a source other than Disclosing Party when the source is not, to the best of Receiving Party's knowledge, subject to a confidentiality obligation to Disclosing Party; (iii) is independently developed by Receiving Party, its representatives or affiliates, without reference to the Confidential Information; (iv) is required to be disclosed by law or valid legal process provided that the Receiving Party intending to make disclosure in response to such requirements or process shall promptly notify the Disclosing Party in advance of such disclosure and reasonably cooperate in attempts to maintain the confidentiality of the Confidential Information.

- (d) As to any individual item of Confidential Information, the restrictions under this Article 25 shall expire five (5) years after the date of disclosure. This Article 25 does not supersede any separate confidentiality or nondisclosure agreement signed by the parties.

26. Non-Survival.

The following Articles shall not survive termination or cancellation of this Agreement: 5, 7, 8, 17 and 18. All other Articles shall survive the termination or cancellation of the Agreement.

27. Entire Agreement.

This Agreement constitutes the entire agreement between ABB and Purchaser. There are no agreements, understandings, restrictions, warranties, or representations between ABB and Purchaser other than those set forth herein or herein provided. As stated in Article 1 of this Agreement, ABB's Proposal, Policies, Addendum(s), if any, submitted to Purchaser, shall control over any conflicting terms. ABB specifically rejects any exceptions to this Agreement, Proposals, Policies, and/or Addendum(s) on the face of any purchase order. Purchaser shall advise ABB in writing of all conflicts, errors, omissions, or discrepancies among the Proposal, Policies, Addendum(s) and this Agreement immediately upon discovery. This Agreement shall supersede any standard, preprinted terms and conditions that are automatically attached to purchase orders issued by Purchaser.

28. US Government Contracts.

This Article 28 applies only if the Equipment, Software, and/or Services are purchased directly or indirectly, with federal, state or local government funding.

- (a) Unless expressly and specifically stated otherwise by ABB herein:

- (i) all Equipment, Software, and/or Services provided by ABB under this Agreement meet the definition of "commercial product," "commercial computer software," "commercial service" and/or "commercial-off-the-shelf item" as defined in FAR Subpart 2.101;
- (ii) consistent with FAR 12.212, all commercial computer software and commercial computer software documentation are acquired by Purchaser under ABB's standard commercial software licenses;
- (iii) to the extent that any domestic preference requirements, whether Buy America Act ("BAA")(41 U.S.C. § 8301 – 8305, as amended) or Build America, Buy America ("BABA")(Pub. L. 117-263, § 58, December 23, 2022,), or Trade Agreements Act of 1979 ("TAA")(Pub. L. 96-39, §1(a), July 26, 1979) or any other, may be applicable to this Agreement, the country of origin of Equipment is unknown, and ABB cannot and shall not be obligated to provide any certification of conformity therewith;
- (iv) there are no prevailing wage requirements attached to this Agreement; that is, (i) any Services provided by ABB are exempt from the Service Contract Labor Standards ("SCA")(41 U.S.C. § 6701 -6707, as amended)(as provided in FAR clause 52.222-41), (ii) any Services provided by ABB for the construction, alteration or repair of public buildings or public works are exempt from the Davis-Bacon Act ("DBA")(40 U.S.C. §§ 3101-9507, as amended) (as provided in FAR clause 52-222-6), and (iii) no state or other local prevailing wage laws apply;
- (v) there are no Security or Safeguarding of Covered Contractor Information Systems requirements as provided by FAR Subpart 4.0404(a) and 4.1903, respectively; that is FAR 52.204-2 and 52.204-21, respectively, do not apply to this Agreement; and
- (vi) any applicable FAR Subpart and/or provision and/or clause referenced in this



Article 28 shall be as of the effective date of this Agreement.

- (b) Purchaser shall notify ABB in writing, prior to the execution of this Agreement, or if this Agreement is a master ordering agreement, prior to ABB's acceptance of any purchase order submitted hereunder, if (1) any of the Services provided hereunder are subject to federal, state or local prevailing wage laws, as described in Article 28 (a)(iv) or (2) the security regulations set forth in Article 28(a)(v) apply. Purchaser shall defend, indemnify and hold ABB harmless from and against all losses, liabilities, cost, expense (including attorney's fees and expenses of litigation and/or settlement), damages, allegations, claims and causes of action arising from Purchaser's failure to notify ABB in writing in accordance with this Article 28(b).
- (c) If Purchaser is an agency of the U.S. Government, then:
 - (i) as permitted by FAR Subpart 12.302, all paragraphs of FAR 52.212-4 (except those listed in 12.302(b)) are replaced with this Agreement's terms and conditions and
 - (ii) all subparagraphs of FAR 52.212-5 apply only to the extent applicable to the sale of COTS and/or commercial items and as appropriate for the Agreement price.
- (d) If Purchaser is procuring the Equipment, Software, and/or Services as a prime contractor or subcontractor at any tier, on behalf of any agency of the U.S. Government, then FAR 52.212-5(e) or 52.244-6 (whichever is applicable) applies only to the extent applicable to the sale of COTS and/or commercial items and as appropriate for the Agreement price.
- (e) For all U.S. federal, state, and/or municipal end-user sales (collectively "U.S. Government"), Purchaser shall disclose and provide all U.S. Government contractual flowdown requirements to ABB when requesting a quote/proposal from or placing a purchase order with ABB. Purchaser shall indemnify, defend, and hold harmless ABB from and against all losses, costs, claims, causes of action, damages, liabilities, and expenses, including attorney's fees,

all expenses of litigation and/or settlement, and court costs, arising from its failure to comply with this obligation.

29. Data Protection.

- (a) The parties agree that the protection of Personal Data is very important. If Purchaser discloses Personal Data to ABB, ABB shall comply with all applicable data protection laws and regulations. Purchaser shall comply with all applicable data protection laws and regulations in respect of any Personal Data it receives from ABB in the course of receiving the Equipment or Services.
- (b) The parties agree that neither will withhold or delay its consent to any changes to this clause which are required to be made in order to comply with applicable data protection laws and regulations and/or with guidelines and order from any competent supervisory authority, and their application to the Equipment or Services from time to time, and agrees to implement any such changes at no additional cost to the other party.
- (c) The parties acknowledge that the processing of Personal Data in accordance with this purchase order may require the conclusion of additional data processing agreements or additional data protection agreements. If and to the extent such additional data processing agreements or additional data protection agreements are not initially concluded as part of the purchase order, the parties shall, and shall ensure that their relevant affiliates or subcontractors shall, upon the other's request promptly enter into any such agreement with an affiliate, as designated by the other party and as required by mandatory law or a competent data protection or other competent authority.

30. Remedies.

The remedies of the Purchaser set forth in this Agreement are exclusive and are its sole remedies for any failure of ABB to comply with its obligations hereunder.